

Machine Learning

CMPSCI 589

Bruno C. da Silva

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An Introduction to ML

UMassAmherst
College of Information
& Computer Sciences



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Welcome Back!

About Me



Bruno Castro da Silva

bsilva@cs.umass.edu

<http://cs.umass.edu/~bsilva>



Autonomous Learning Laboratory

UMass & CICS are here to help

Concerns / Questions / Assistance / Help ?

You can always talk to me

Emma Anderson



Erika Dawson Head



UMass & CICS are here to help

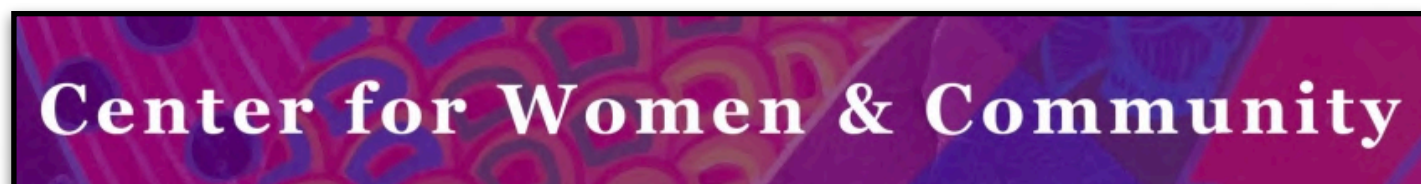


<https://www.umass.edu/studentlife/bewellumass>

UMassAmherst

Center for Counseling
and Psychological Health

<https://www.umass.edu/counseling/>



<https://www.umass.edu/cwc/>

UMassAmherst

Disability Services

<https://www.umass.edu/disability/>

Machine Learning

CMPSCI 589

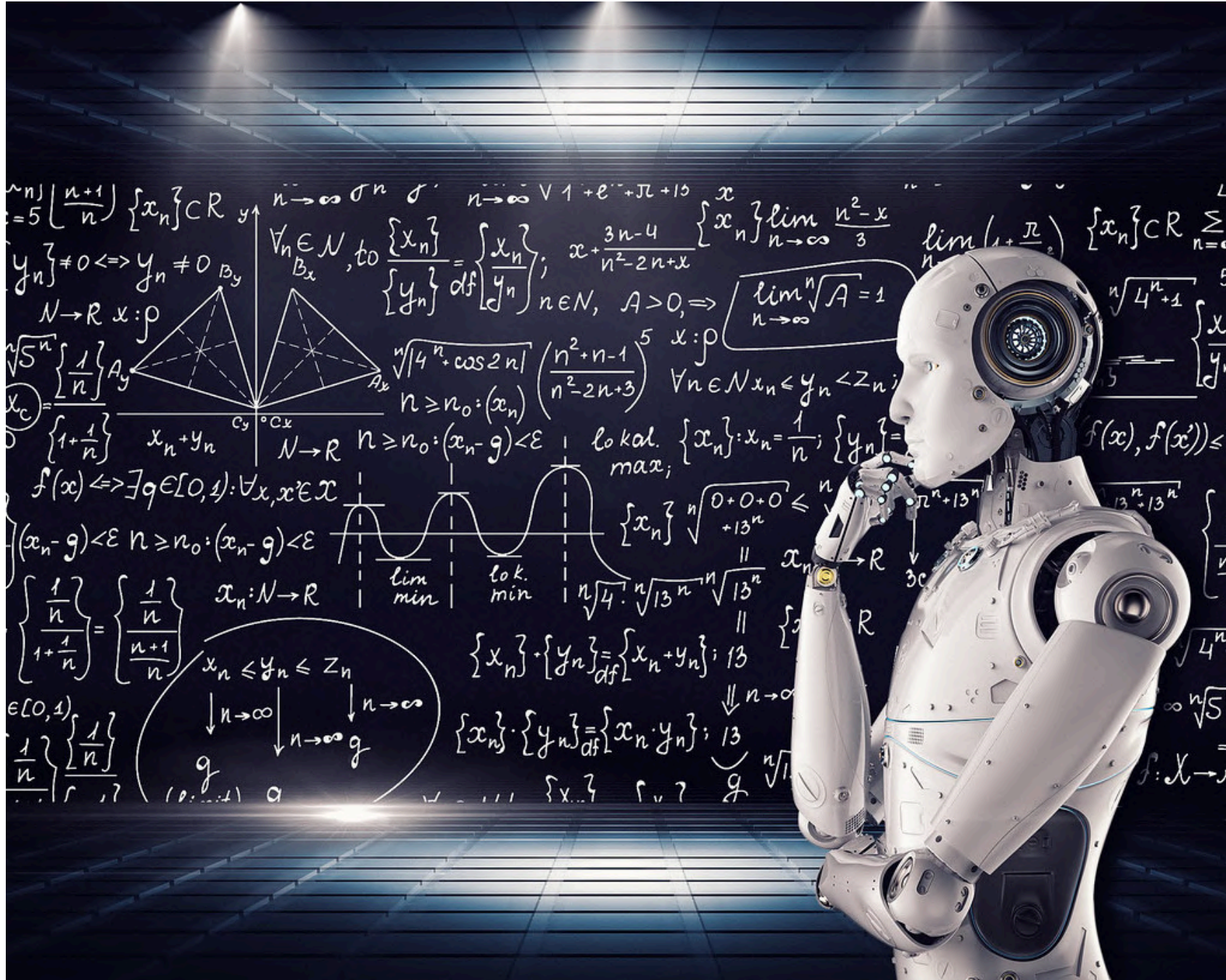


Photo: Mike MacKenzie

Machine Learning

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An Extremely High-Level Introduction to ML

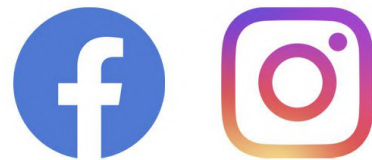
Machine Learning



Google

Meta

Bank of America



amazon

ebay

NETFLIX



You Tube

Uber

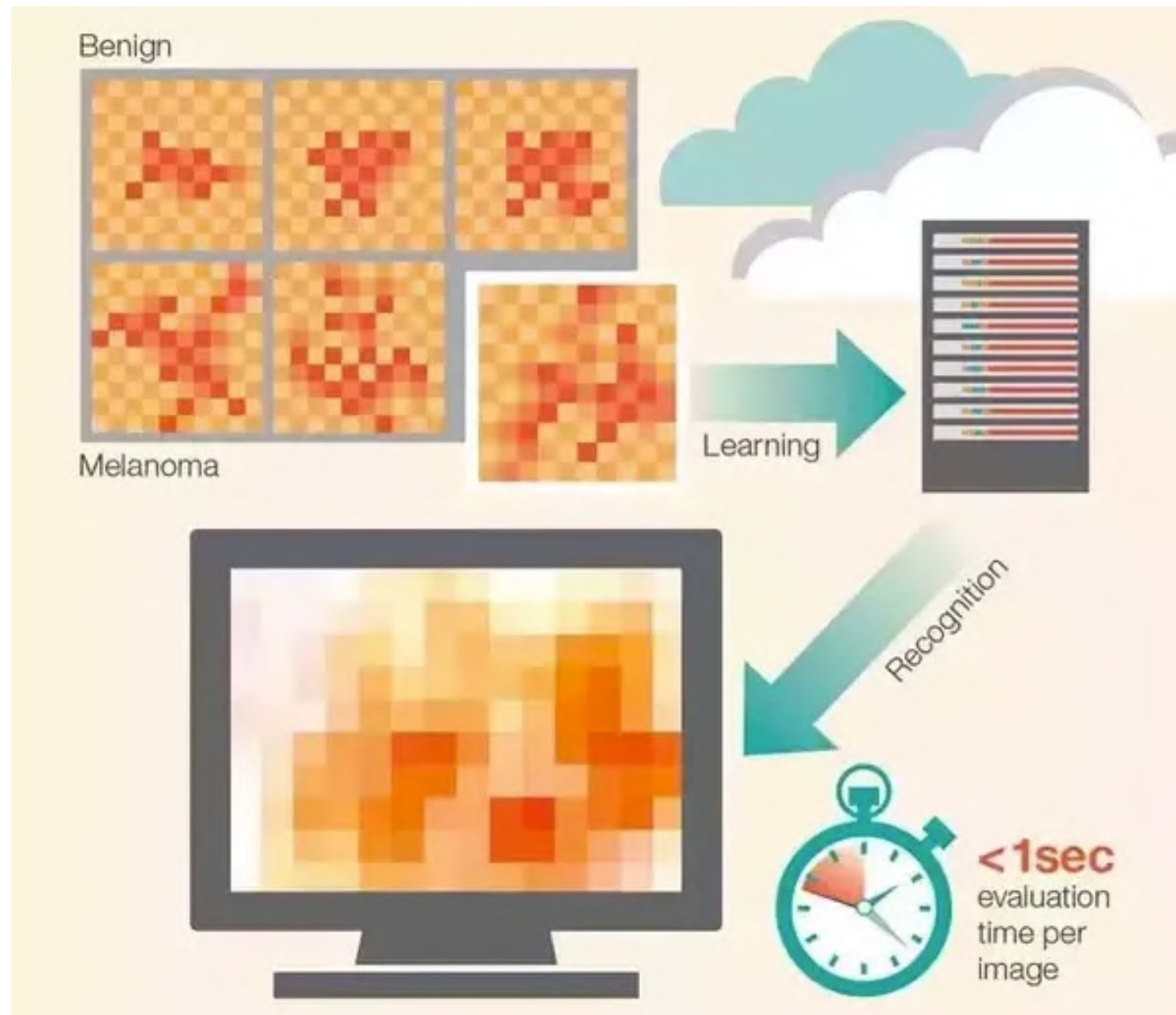
A Few Current Applications of ML



(Siri, Google Assistant, Alexa, etc)

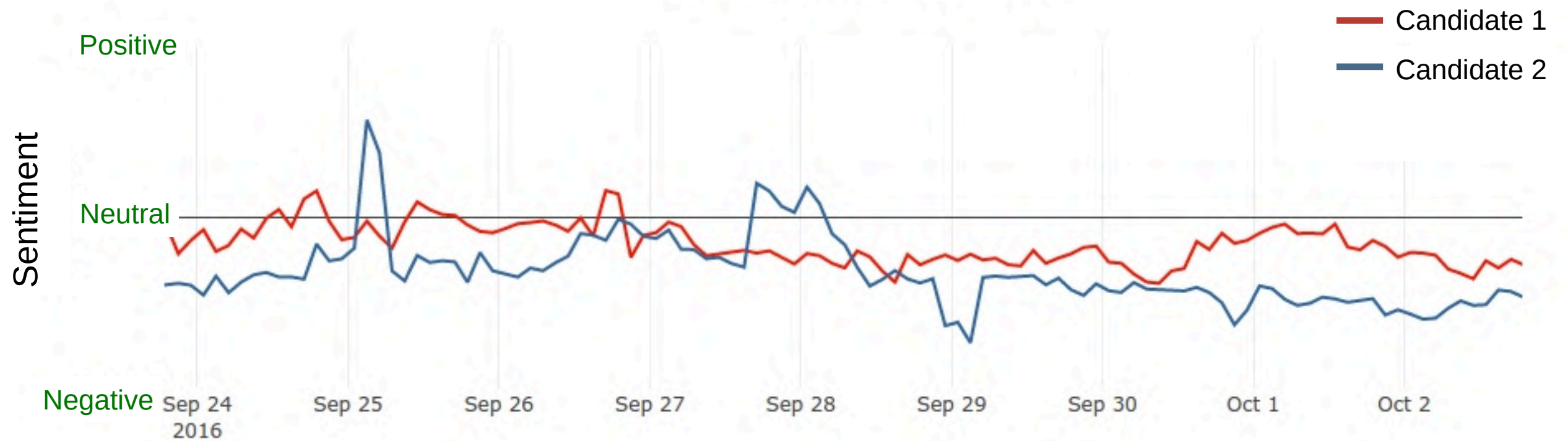
Natural Language Processing & Understanding

“IBM detects skin cancer more quickly with visual machine learning”



Medical Diagnosis

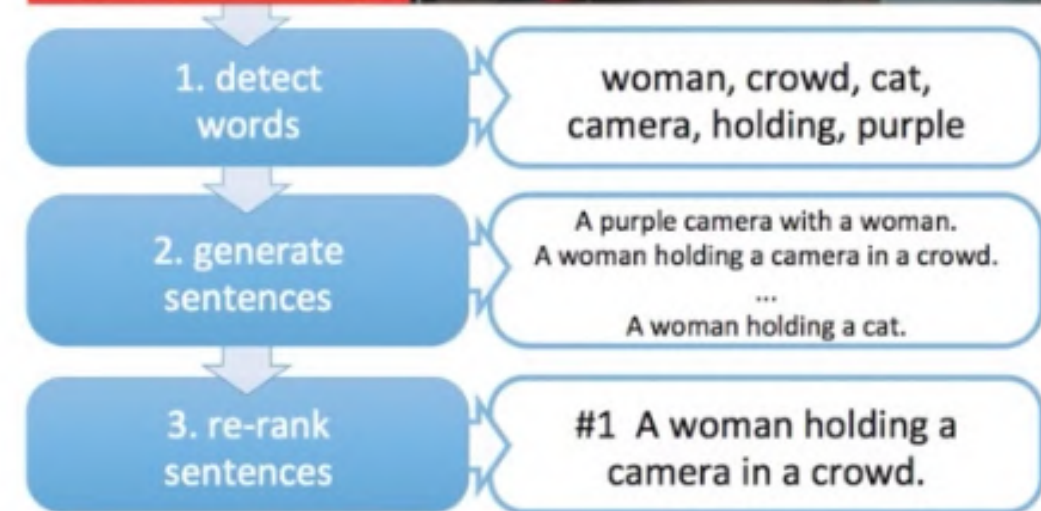
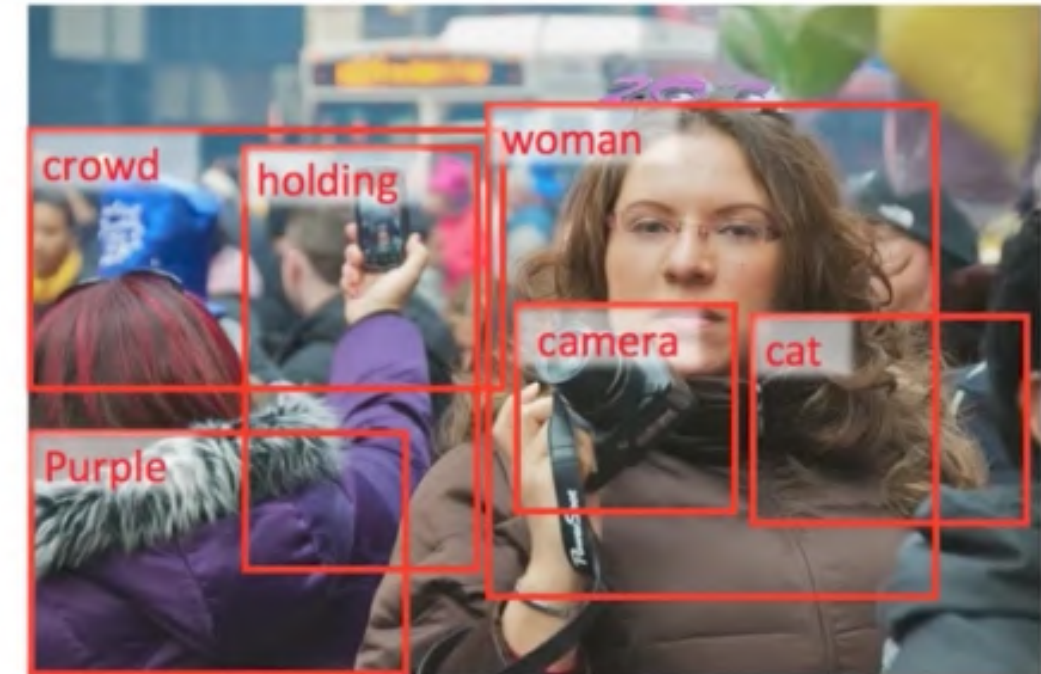
Candidate sentiment over time



Sentiment Analysis



a man sitting on a couch with a dog
 a man sitting on a chair with a dog in his lap



Scene Understanding & Description

NETFLIX

Search

Because you watched shows about Anti-Heroes and Moral Ambiguity >



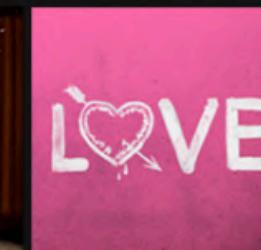
Because you watched shows with Sharp Humor and Strong Female Leads >



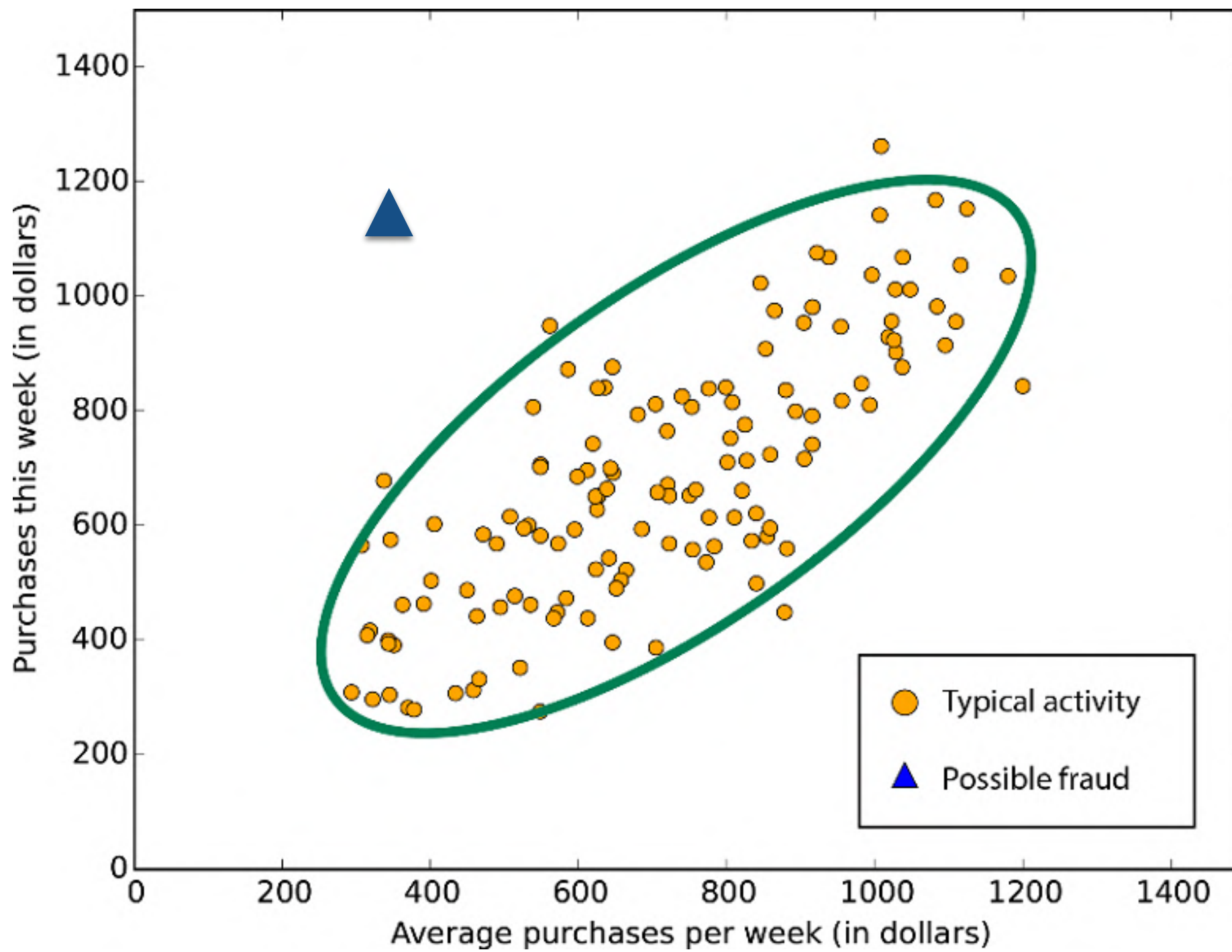
Because you watched shows about Dangerous Worlds and Complex Consequences >



Because you watched shows about Edgy Coming of Age Tales >



Learning User Preferences & Interests



Fraud Detection



IDEAFEED

Google's Self-Driving Cars Are Ridiculously Safe

Autonomous Cars

How I made \$500k with machine learning and HFT (high frequency trading)

Nov 6, 2012

This post will detail what I did to make approx. 500k from high frequency trading from 2009 to 2010. Since I was trading completely independently and am no longer running my program I'm happy to tell all. My trading was mostly in Russel 2000 and DAX futures contracts.

The key to my success, I believe, was not in a sophisticated financial equation but rather in the overall algorithm design which tied together many simple components and used machine learning to optimize for maximum profitability. You won't need to know any sophisticated terminology here because when I setup my program it was all based on intuition. (Andrew Ng's amazing [machine learning course](#) was not yet available - btw if you click that link you'll be taken to my current project: CourseTalk, a review site for MOOCs)

First, I just want to demonstrate that my success was not simply the result of luck. My program made 1000-4000 trades per day (half long, half short) and never got into positions of more than a few contracts at a time. This meant the random luck from any one particular trade averaged out pretty fast. The result was I never lost more than \$2000 in one day and never had a losing month:

Monthly P&L



Hi, I'm Jesse, founder of Thinklab. I live and play in San Francisco. You've found my home on the web.. Welcome!



Tweets

Follow

Financial Services & Applications

Write a short two-paragraph poem about why chocolate is objectively the best food ever. 

Chocolate, oh chocolate, the taste of pure bliss,
A treat that's indulgent, with every bite, a kiss.
From its rich and creamy texture, to its velvety smooth,
It's a flavor that's simply unmatched, a timeless proof.
With notes of sweet and bitter, a balance that's divine,
Chocolate is the food of gods, it's simply so fine.

With a history that dates back, to the ancient Mayans,
Its popularity endures, across cultures and spans.
From a simple bar to elaborate desserts so grand,
Chocolate always steals the show, it's simply unparalleled.
So let's celebrate this food, that brings so much joy,
Chocolate is the best, there's simply no other choice, oh boy!



ChatGPT — Large Language Models



"Teddy bears working on new AI research on the moon in the 1980s"

DALL-E — Realistic Image Creation from Natural Language

Machine Learning

Three Machine Learning approaches

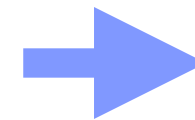
- **Supervised Learning**
- Unsupervised Learning
- Reinforcement Learning

Supervised Learning

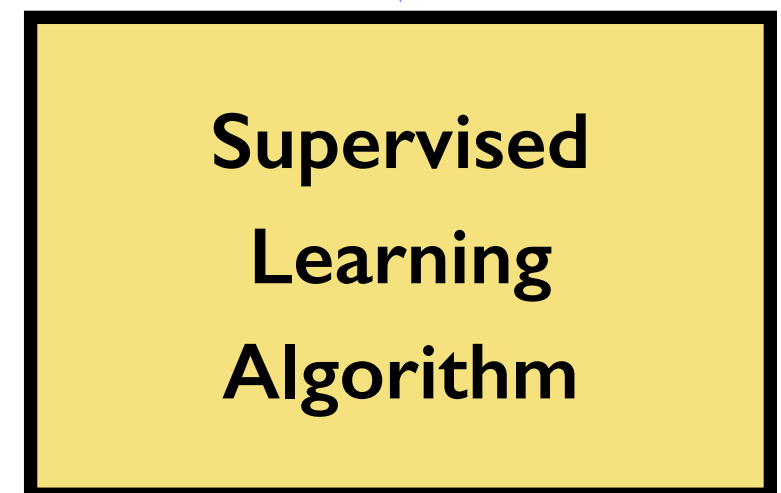
A dataset containing *labeled training examples* is given to the AI

Training Dataset

{info_client_1, Repaid_Loan}
{info_client_2, Did_Not_Repay_Loan}
{info_client_3, Did_Not_Repay_Loan}
....
{info_client_i, Repaid_Loan}
....
{info_client_N, Did_Not_Repay_Loan}



{info_new_client, ?}



{info_new_client, Will_Not_Repay_Loan}

Supervised Learning

A dataset containing labeled training examples is given to the AI

Training Dataset

- Goal is to learn a **predictive model**
- based on training data, should predict the value of some attribute
- *"given all information about patient X, will they have cancer?"*

{info_client_1, Repaid_Loan}
{info_client_2, Did_Not_Repay_Loan}
{info_client_3, Did_Not_Repay_Loan}
....
{info_client_i, Repaid_Loan}
....
{info_client_N, Did_Not_Repay_Loan}

{info_new_client, ?}

Supervised
Learning
Algorithm

{info_new_client, Will_Not_Repay_Loan}

Machine Learning

Three Machine Learning approaches

- Supervised Learning
- **Unsupervised Learning**
- Reinforcement Learning

Unsupervised Learning

A dataset containing unlabeled data is given to the AI
(no “correct/expected” answer associated with each example)

Training Dataset

```
{fuelEconomy=1, lowPrice=3, comfort=5, acceleration=1}  
{fuelEconomy=3, lowPrice=5, comfort=3, acceleration=3}  
{fuelEconomy=5, lowPrice=5, comfort=2, acceleration=4}  
{fuelEconomy=2, lowPrice=2, comfort=1, acceleration=4}  
{fuelEconomy=1, lowPrice=3, comfort=2, acceleration=2}
```

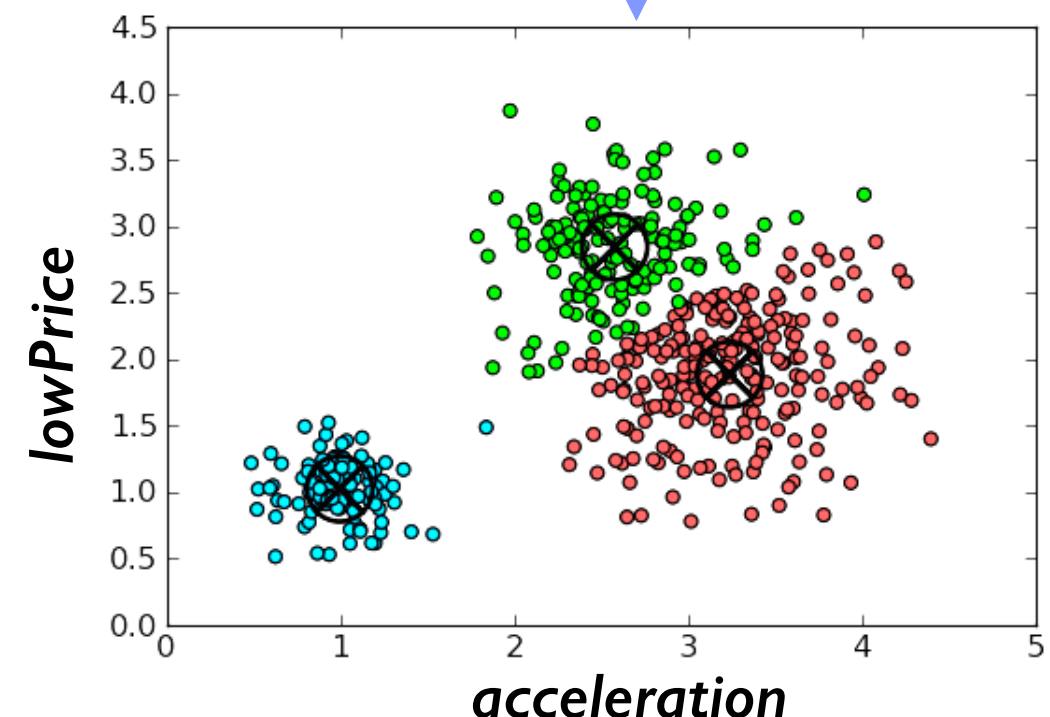
(features buyers said were important when making a decision)

“luxury car buyers”

“car buyers interested in performance/speed”

“car buyers who prioritize economy and comfort”

Unsupervised
Learning
Algorithm



Unsupervised Learning

A dataset containing unlabeled data is given to the AI
(no “correct/expected” answer associated with each example)

Training Dataset

- Goal is to learn a **descriptive model**
 - based on training data, **find patterns or regularities**
 - describe and explore available data

(features buyers said were important when making a decision)

“luxury car buyers”

“car buyers interested in performance/speed”

“car buyers who prioritize economy and comfort”

Unsupervised Learning Algorithm



Machine Learning

- Supervised Learning → there is a dataset with examples of which decisions you should make, depending on the situation
- Unsupervised Learning → there is no dataset with examples. Just lots of data, and we need to find interesting patterns
- Reinforcement Learning → there is someone evaluating your actions, but not telling you *which* actions would have been best/correct

Machine Learning

- **Supervised Learning** → there is a dataset with examples of which decisions you should make, depending on the situation
- **Unsupervised Learning** → there is a dataset with examples. Just lots of data, interesting patterns

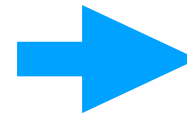
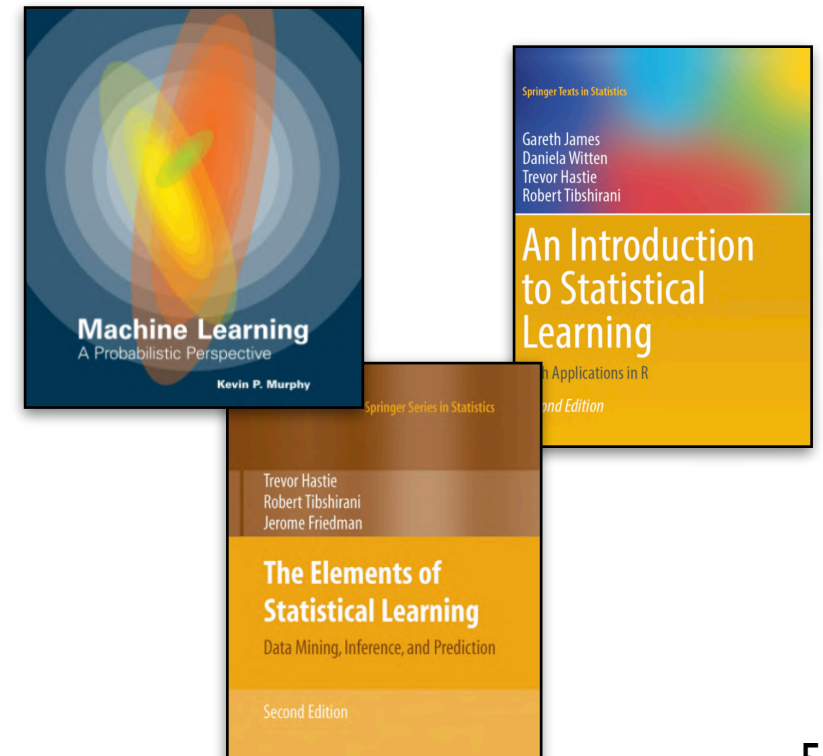
Wrist angle
should have been
13 degrees



Intermission: Syllabus

Syllabus

- **Website:** cs.umass.edu/~bsilva [look for "Teaching"]
- **Canvas:**
 - Syllabus
 - Lectures will be recorded and made available on Canvas
 - Slides
 - Homework & Assignments
- **Textbooks:**
 - No textbooks are required
 - A few options are available for free online
 - Link on Canvas



Syllabus

- **CampusWire:** *Link on Canvas*
- **Teaching Assistants:** *See syllabus for contact information*



Ke Xiao



Sunjae Kwon



Dhawal Gupta



Nana Sereboo



John Raisbeck

Syllabus

- **CampusWire:** *Link on Canvas*
- **Teaching Assistants:** *See syllabus for contact information*

We will hold office hours every day

See syllabus for schedule and more details

Syllabus

Your grade will have three components:

1. **Homework Assignments (50%)**: There will be frequent homework assignments, both written and programming. All assignments will have an equal weight.
2. **Midterm exam (30%)**.
3. **Project (20%)**: Given the wide range of possible real-life applications of machine learning, it is critical that we study how to implement, fine-tune, and deploy these algorithms in practice. Further details will be available when the project is assigned, after the most relevant course material has been covered.

[90% - 100%]	→	A-	A	
[75% - 90%)	→	B-	B	B+
[65% - 75%)	→	C-	C	C+
[55% - 65%)	→	D	D+	

Course grades will always (and only) be curved in student's favor

Syllabus

- **Late Policy**

- Deadlines in this course are **strict**. A submission one minute after the deadline will receive zero credit. You are strongly encouraged to submit hours before any deadline.
- To allow some flexibility to complete assignments (*homeworks*) given other constraints, you have a total of seven free late days (...) After you have used up your late days, late homework will not count for credit except in special circumstances (i.e., illness documented by a doctor's note).
- All exams must be taken at the scheduled time unless (1) there is a documented conflict and arrangements have been made with the instructor before the exam; or (2) you have a medical emergency and you bring proof of such to the instructor before final grades for the given exam are computed. In any other case (unless those covered by [University's Academic Regulations](#)), missing an exam will result in a grade of "F" for that exam.

Syllabus

- **Disability Services**

The University of Massachusetts is committed to providing an equal educational opportunity for all students. If you have a documented physical, psychological, or learning disability on file with **Disability Services**, you may be eligible for academic accommodations to help you succeed in this course. If you would like to register with Disability Services, please visit their **website** or their office (161 Whitmore Administration Building; phone (413) 545-0892). Finally, if you have a documented disability that requires an accommodation, **please notify me within the first two weeks of the semester so that we can make appropriate arrangements.**



<https://www.umass.edu/disability/>

Syllabus

- **Cheating**

- Cheating will not be tolerated. Assignments may include instructions about what forms of collaboration are allowed, if/when relevant.
- Copying answers or code from external sources (books, web pages, etc.), from other students, or from solutions to assignments from previous years is always considered cheating. Note that, according to the new UMass Academic Honesty Policy, the use of AI text generators (such as **ChatGPT**) is prohibited. *To emphasize: no detectable copying is acceptable, even, e.g., copying a single sentence from an outside source. Sharing your code or solutions with other students is also considered cheating.*
- (...)
- **All instances of cheating will be reported to the university's Academic Honesty Board.** Any detected cheating will result either (i) in a grade of -100% on the assignment for all students involved (negative credit); or (ii) a grade of **F** in the course. The instructor will decide at their discretion which of these possible resolutions is more appropriate.

Syllabus

- Cheating

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- Historically:

- Students who failed directly (**F**) ~13.7%
- Students caught (*other penalties*): ~24.2%

Supervised Learning

Supervised Learning

- Predict number of **Likes** an Instagram post/picture might get
- Based on
 - how many friends were online
 - how funny the user believes the post is (0-10)



	#friends	funny score	Likes
Post1	40	8	28
Post2	36	10	28
Post3	20	6	16
Post4	56	4	31
Post5	58	0	29
Post6	46	10	33

(<http://qr.ae/TbcC8T>)

Supervised Learning

	#friends	funny score	Likes	
Post1	40	8	28	28
Post2	36	10	28	28
Post3	20	6	16	16
Post4	56	4	31	32
Post5	58	0	29	29
Post6	46	10	33	33



It seems like
there is a *pattern* here!

$$\text{Likes} = (0.5 * \text{\#friends}) + (1 * \text{funny_score})$$

Machine learning algorithms learn these "weights"

Supervised Learning

	#friends	funny score	Likes	
Post1	40	8	28	28
Post2	36	10	28	28
Post3	20	6	16	16
Post4	56	4	31	32
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It seems like
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$$\text{Likes} = (0.5 * \text{\#friends}) + (1 * \text{funny_score})$$

Linear Regression

Supervised Learning

	#friends	funny score	Likes	
Post1	40	8	28	28
Post2	36	10	28	28
Post3	20	6	16	16
Post4	56	4	31	32
Post5	58	0	29	29
Post6	46	10	33	33



It seems like
there is a *pattern* here!

Likes = complicatedFunction(**#friends**, funny_score)

Neural Networks

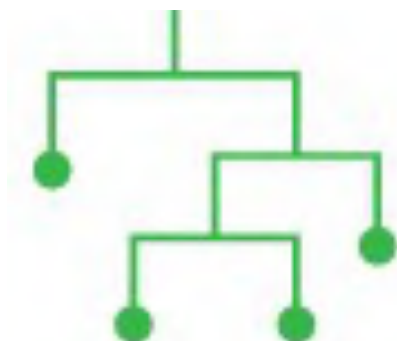
Supervised Learning

	#friends	funny score	Likes
Post1	40	8	28
Post2	36	10	28
Post3	20	6	16
Post4	56	4	31
Post5	58	0	29
Post6	46	10	33



```
if #friends >= 50
  if funny_score > 3
    Likes >= 30
  else
    Likes < 30
else
  if funny_score > 9
    Likes >= 30
  else
    Likes < 30
```

Decision Trees



Learning to Play the 20 Questions Game

<http://en.akinator.com/>

Question Nº 1

Is your character a female?

Yes

No

Don't know

Probably

Probably not



Correct

Question Nº 2

Is your character real?

Yes

No

Don't know

Probably

Probably not



Correct

Question Nº 3

Is your character a famous
youtuber?

Yes

No

Don't know

Probably

Probably not



Correct

Question Nº 4

Is your character older than 35 years old?

Yes

No

Don't know

Probably

Probably not



Correct

Question Nº 5

Is your character American ?

Yes

No

Don't know

Probably

Probably not



Correct

Question Nº6

Is your character an actor?

Yes

No

Don't know

Probably

Probably not



Correct

Question Nº 7

Is your character the president
or was he?

Yes

No

Don't know

Probably

Probably not



Correct

Question Nº 8

Is your character black?

Yes

No

Don't know

Probably

Probably not





I think of
Barack Obama

Former President of the United States

© Copyright / IP Policy

Yes

No

Learning to Play the 20 Questions Game

<http://en.akinator.com/>

Training Dataset

{man=True, American=True, Politician=False, Singer=True, Dead=True}	→ Elvis
{man=False, American=False, Politician=True, Singer=False, Dead=False}	→ Angela Merkel
{man=True, American=True, Politician=True, Singer=False, Dead=False}	→ Obama
{man=True, American=False, Politician=False, Singer=True, Dead=False}	→ Bob Marley

Decision Trees

- Which attributes of a person to check first, to guess as fast as possible?
 - Is the person a man or a woman?
 - Is the person older than 5 years old?

Information Gain

- $G(\text{"Gender"}) = 0.9$
- $G(\text{"Lives_in_USA"}) = 0.73$
- $G(\text{"Is_Politician"}) = 0.36$
- ...

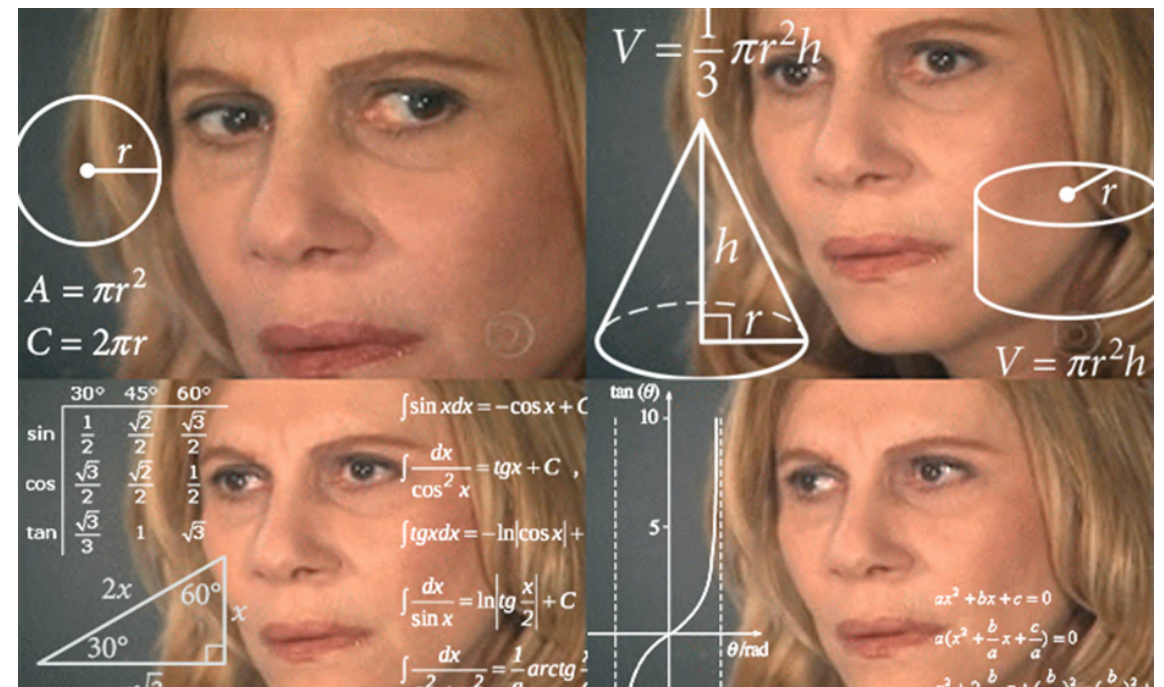
Entropy of a set (or dataset)

Decision Trees

- Which attributes of a person to check first, to guess as fast as possible?
 - Is the person a man or a woman?
 - Is the person older than 5 years old?

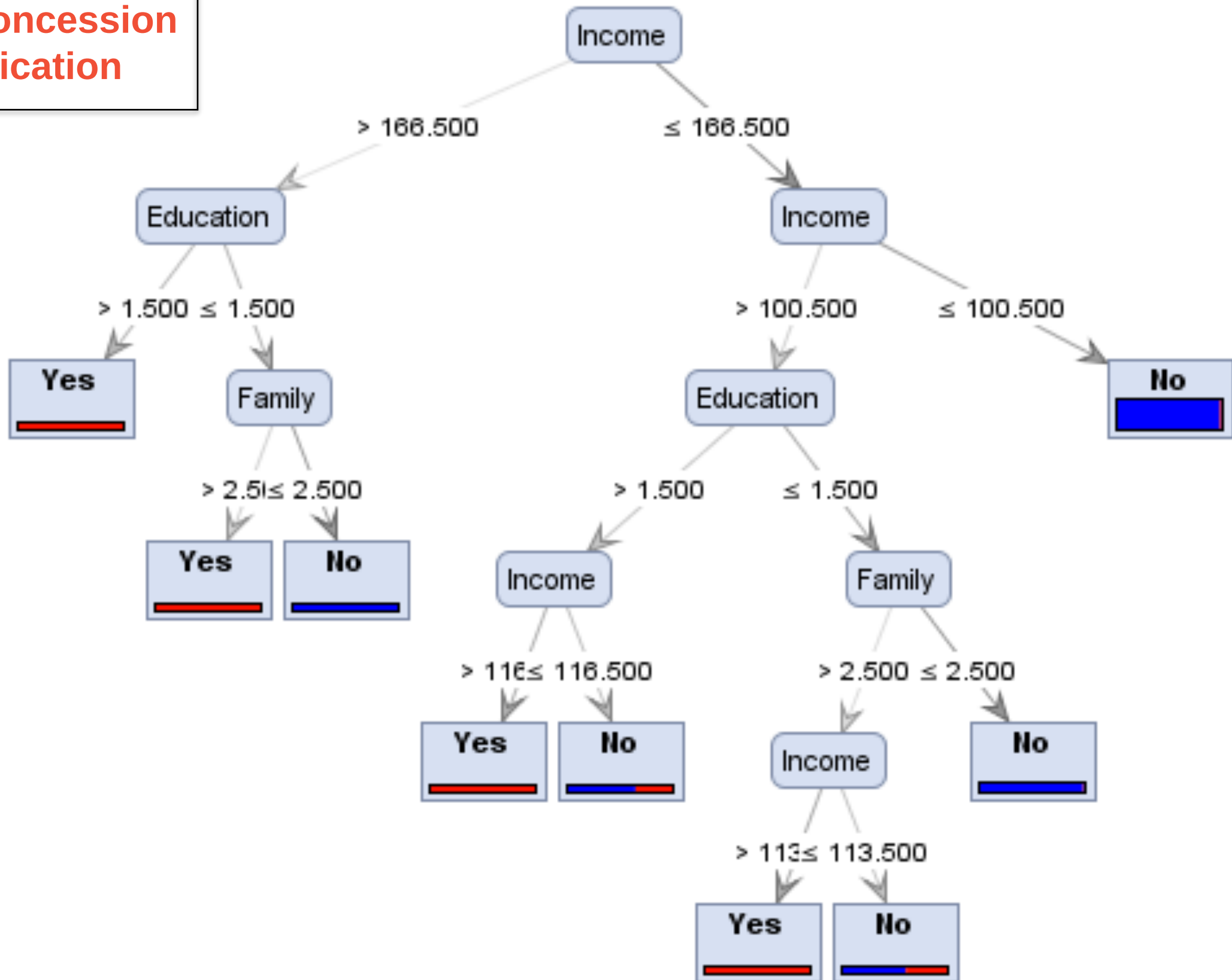
Information Gain

- $G(\text{"Gender"}) = 0.9$
- $G(\text{"Lives_in_USA"}) = 0.73$
- $G(\text{"Is_Politician"}) = 0.36$
- ...



Decision Trees

Loan Concession Application



Neural Networks

- Pattern recognition
 - image processing, facial recognition, vision for robotics
 - speech recognition
 - text classification
 - machine translation
 - super-human performance in videogames
 - weather forecasting
 - etc.

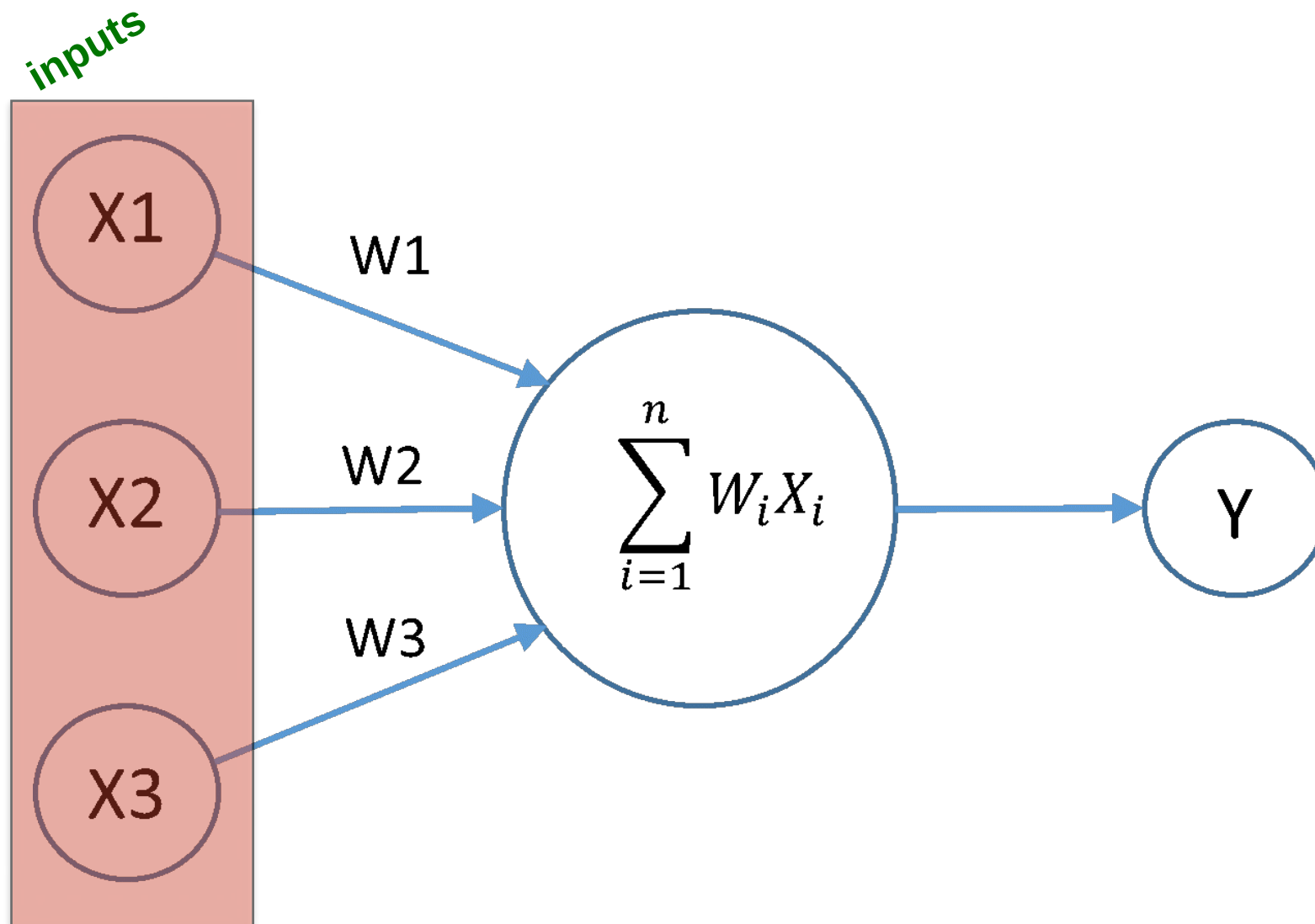
Neural Networks

Tesla Auto-Pilot Saves the Day

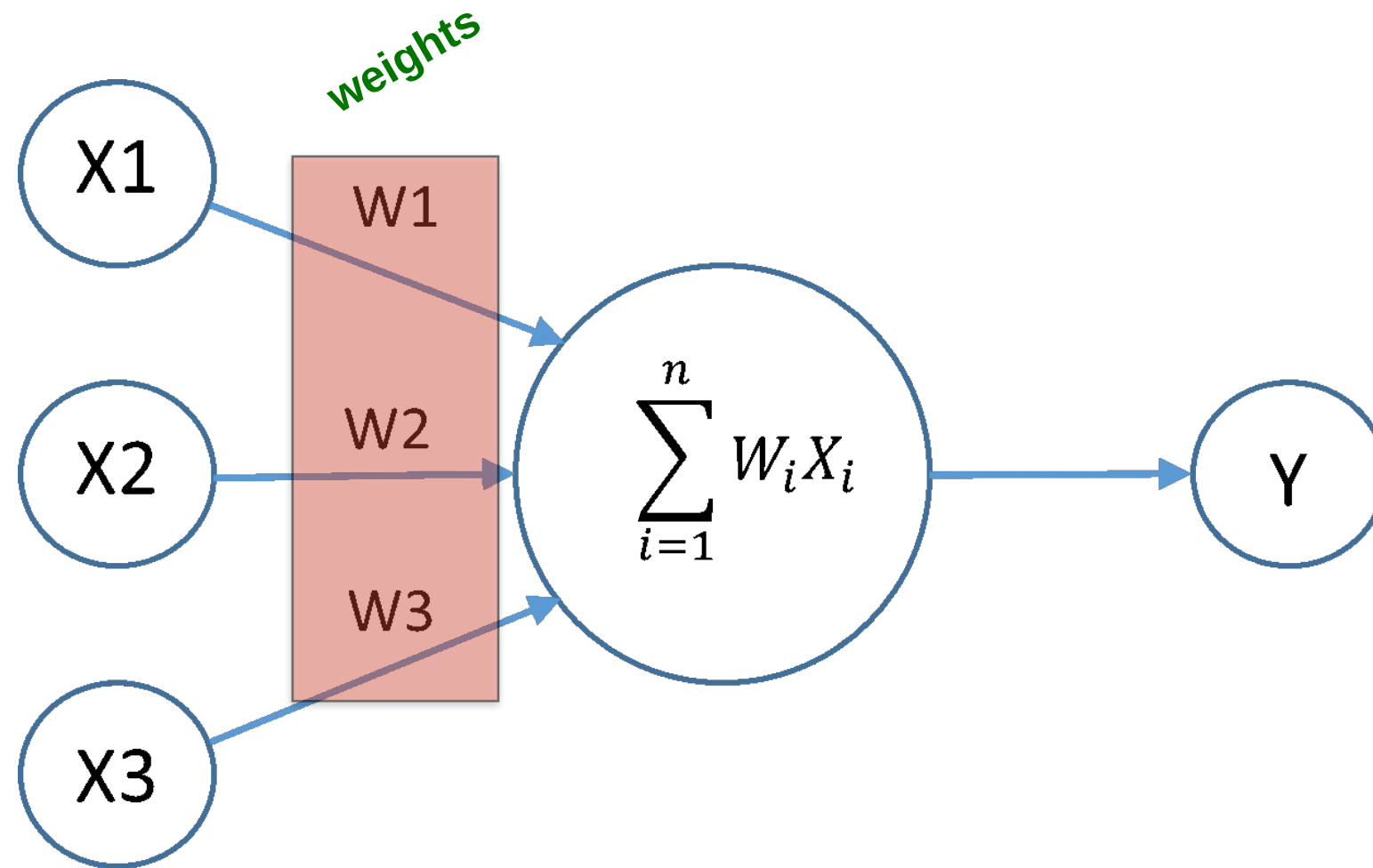


<https://www.youtube.com/watch?v=9X-5fKzmy38>

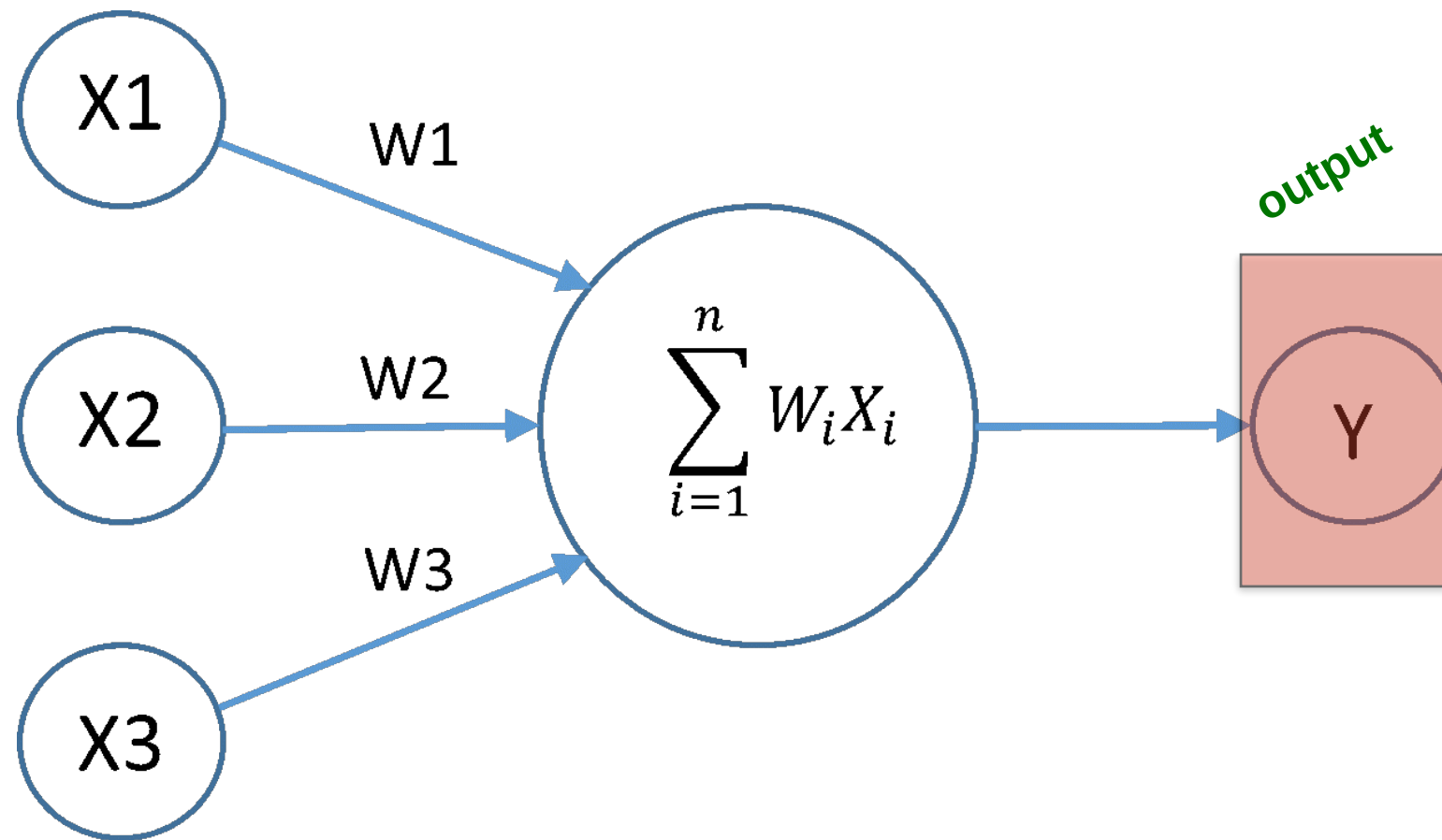
A Simple Model of a Neuron



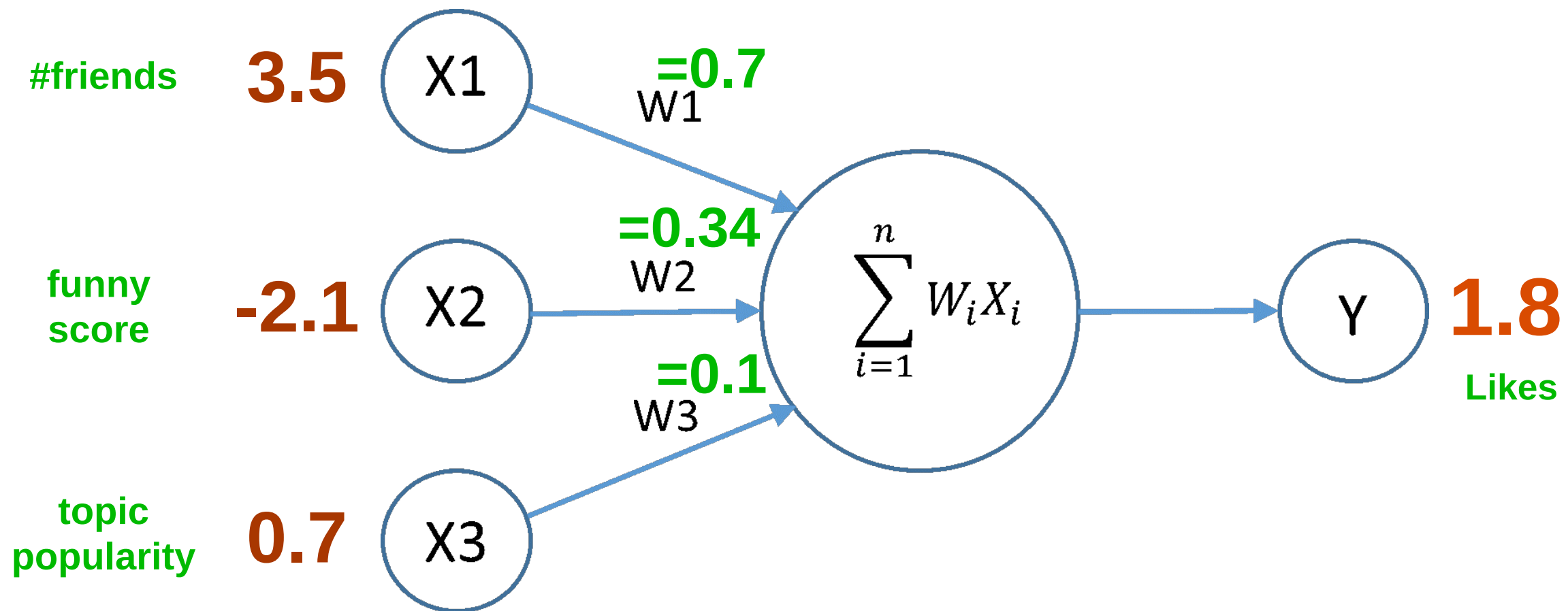
A Simple Model of a Neuron



A Simple Model of a Neuron

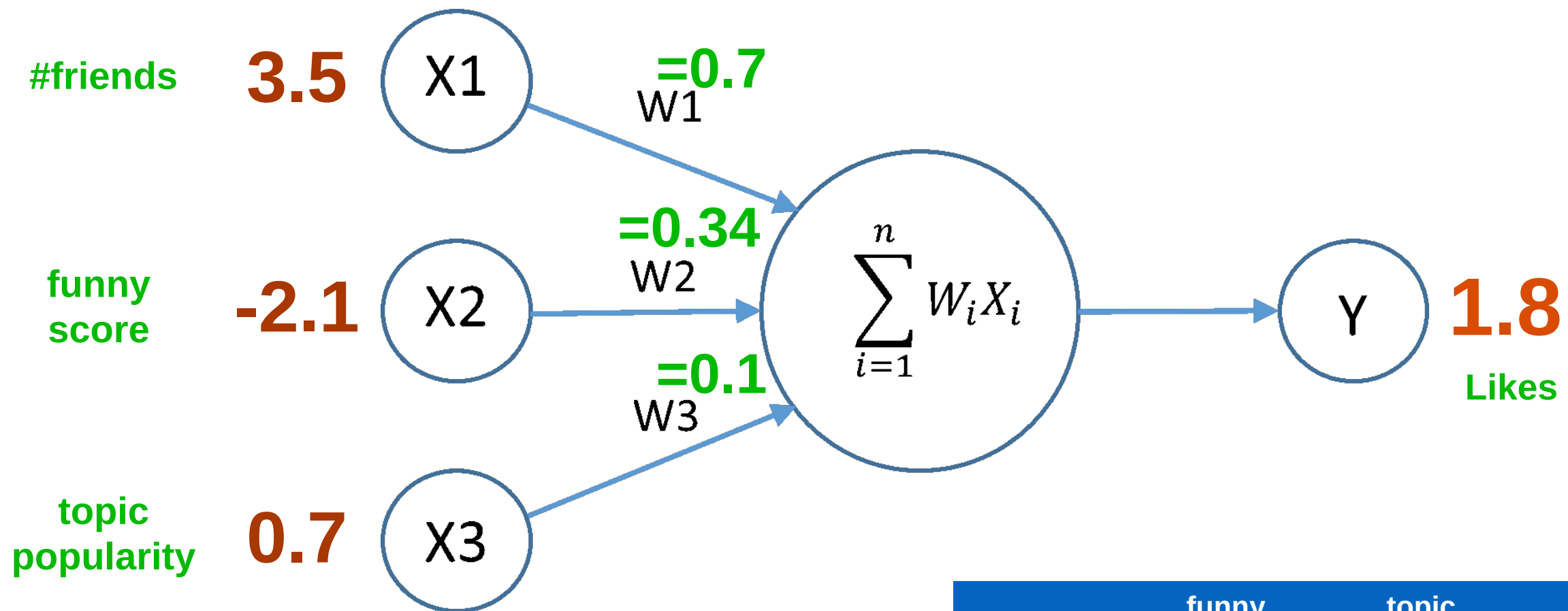


A Simple Model of a Neuron



$$3.5 * 0.7 - 2.1 * 0.34 + 0.7 * 0.1 = 1.8$$

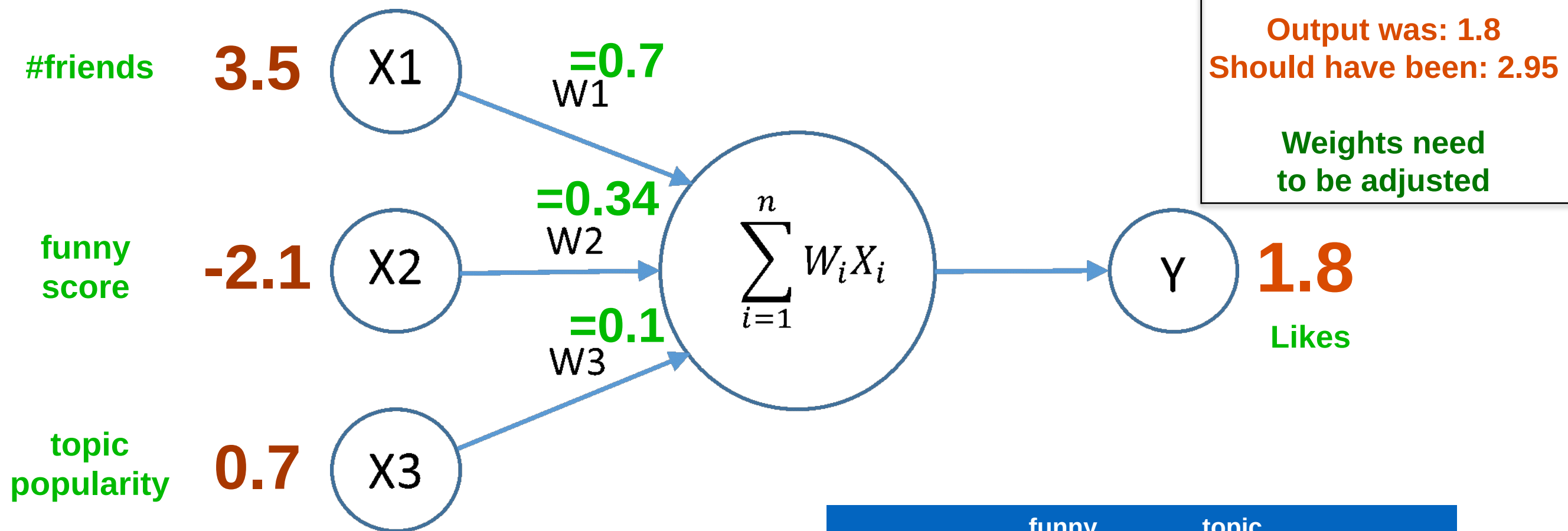
A Simple Model of a Neuron



#friends	funny score	topic popularity	Likes
1.0	2.4	0.8	25
45.2	9.1	2.4	126
3.5	-2.1	0.7	2.95
0	113	45	25.6

Where do weights come from?
Training data

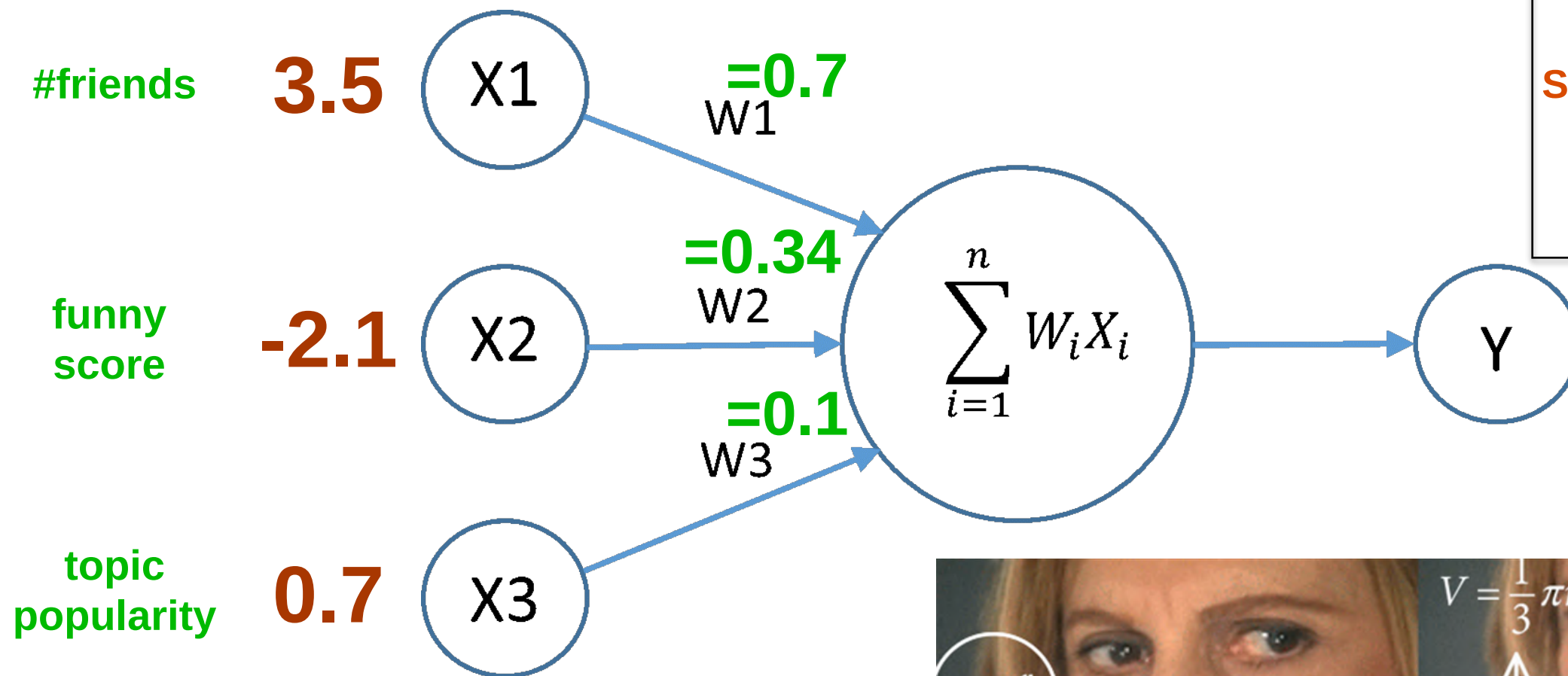
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#friends	funny score	topic popularity	Likes
1.0	2.4	0.8	25
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A Simple Model of a Neuron

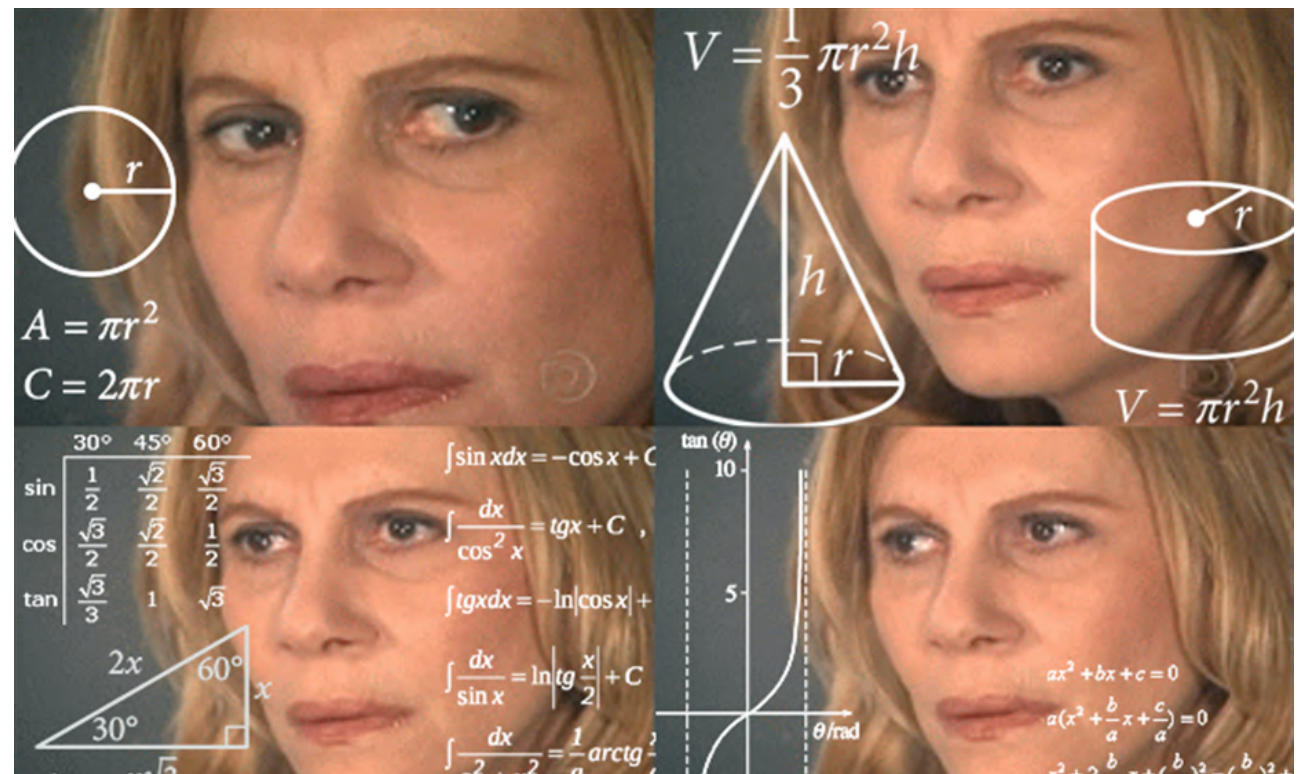


Network's prediction is wrong!

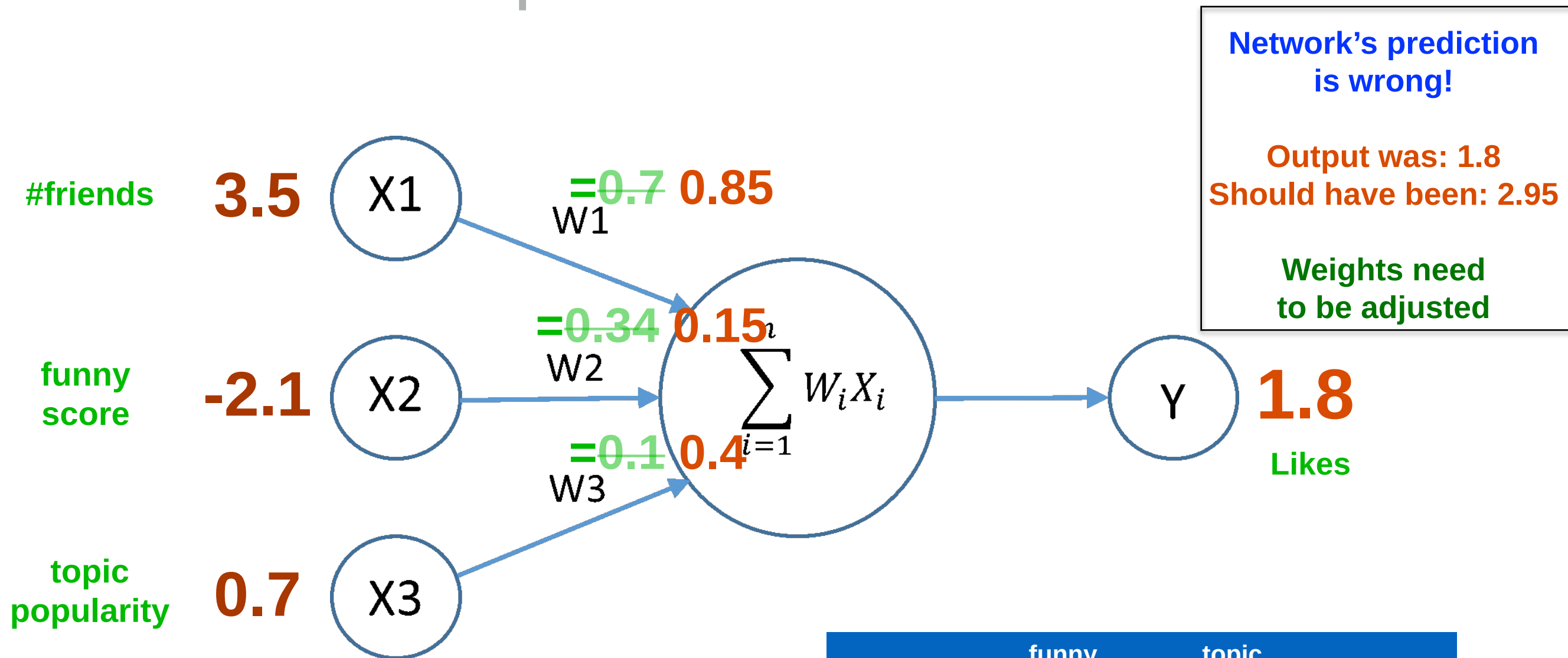
Output was: 1.8
Should have been: 2.95

Weights need to be adjusted

Where do weights come from?
Training data



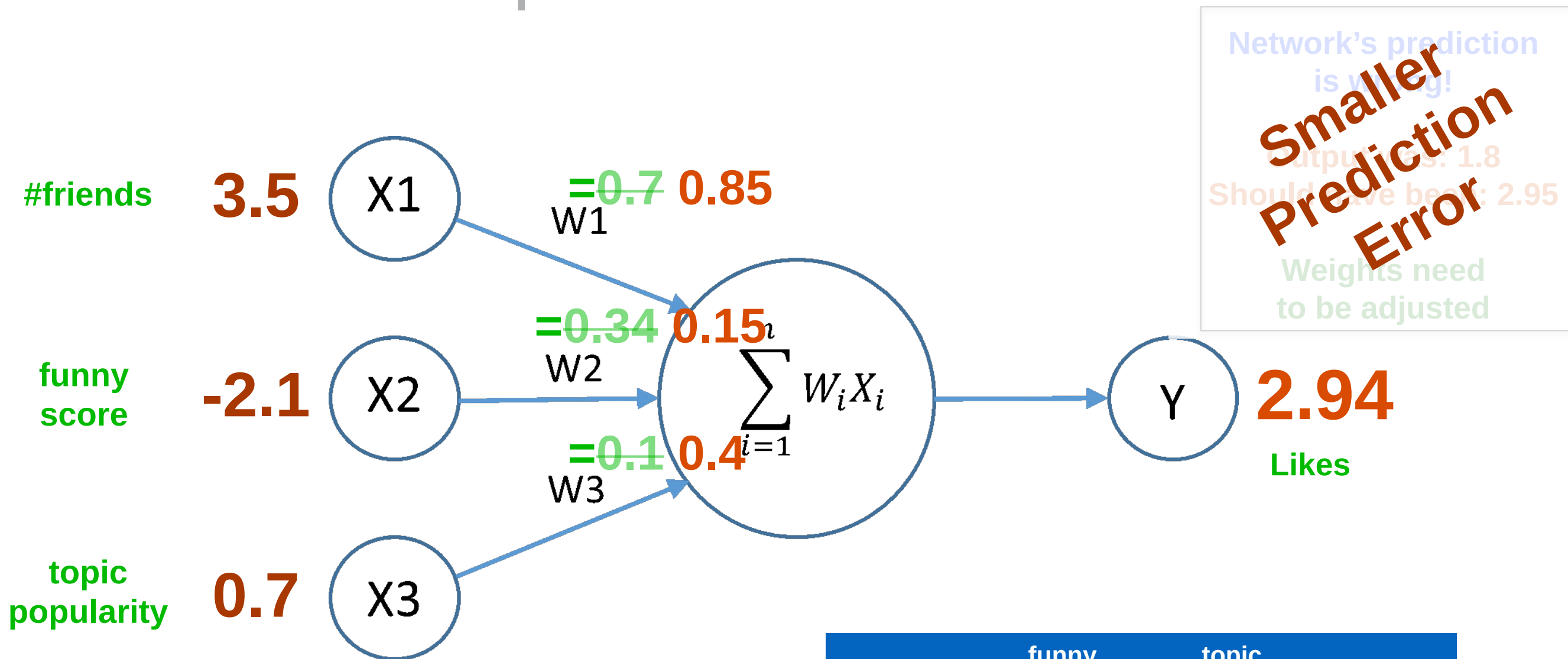
A Simple Model of a Neuron



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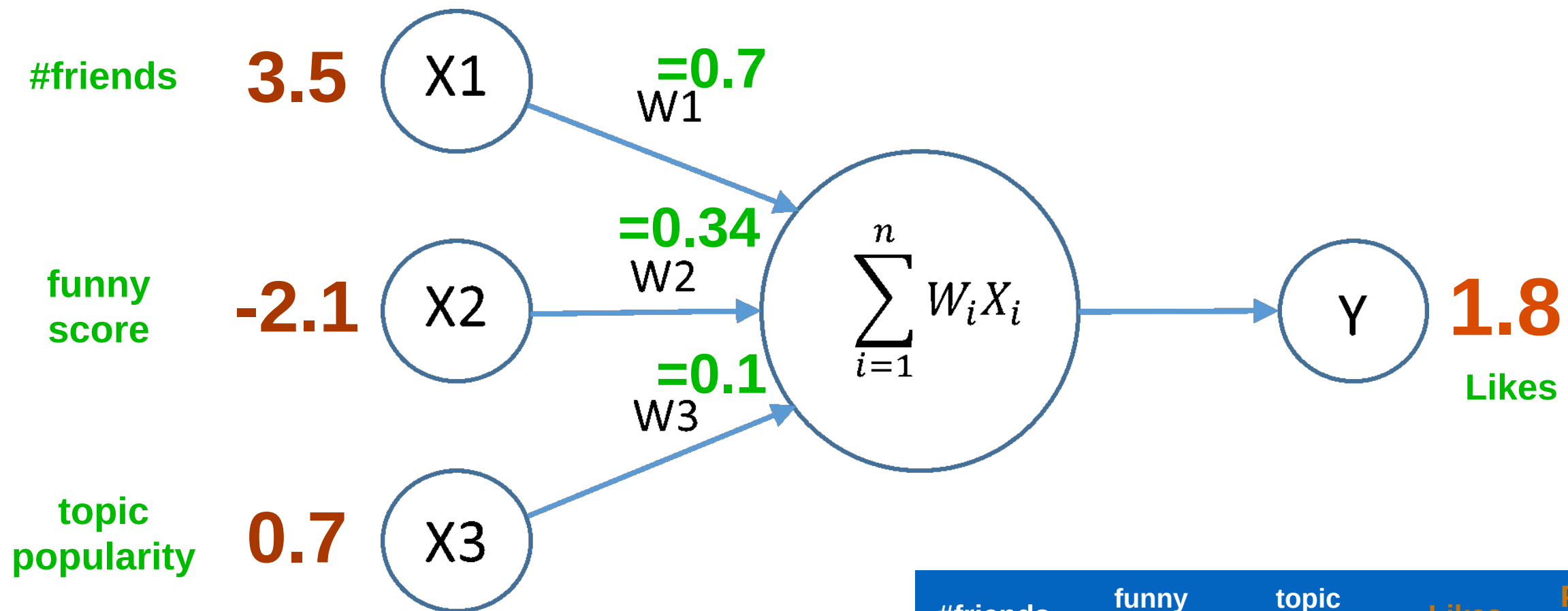
A Simple Model of a Neuron



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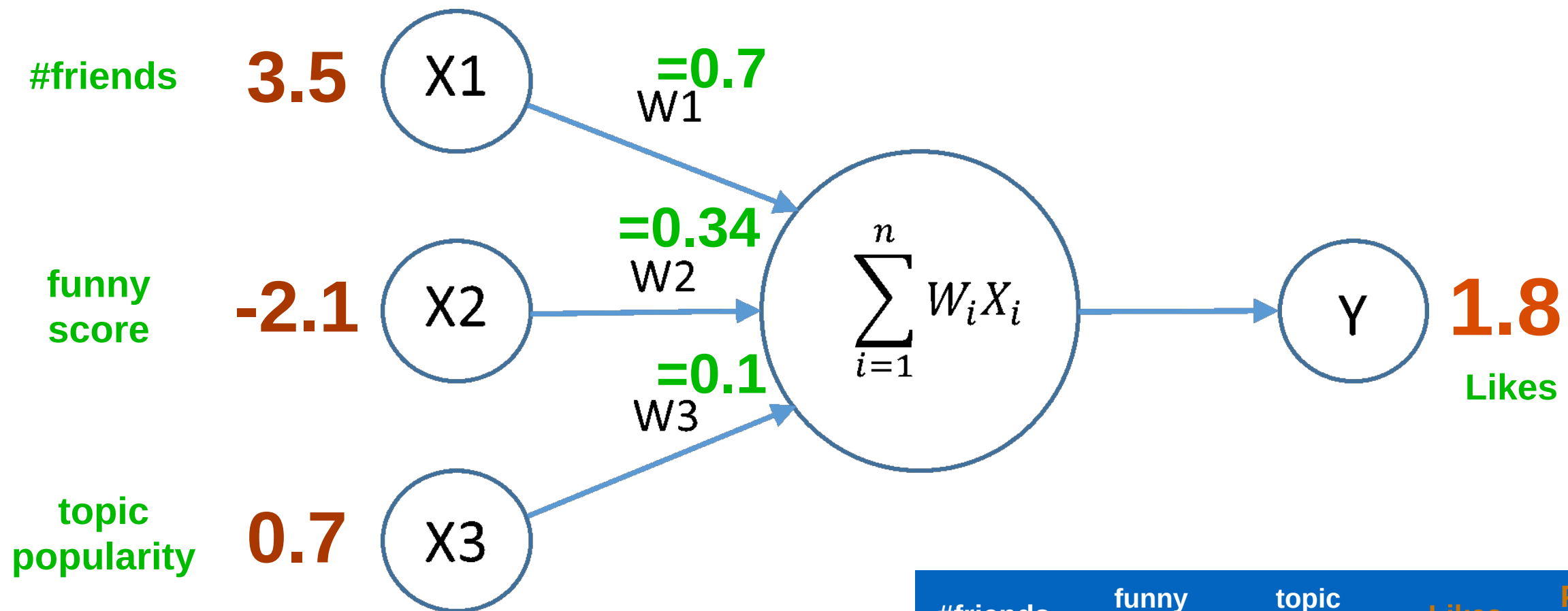


Tests example
#1

Adjusts weights
based on the error
(25-21)

#friends	funny score	topic popularity	Likes	Prediction Likes
1.0	2.4	0.8	25	21
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3.5	-2.1	0.7	2.95	
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A Simple Model of a Neuron

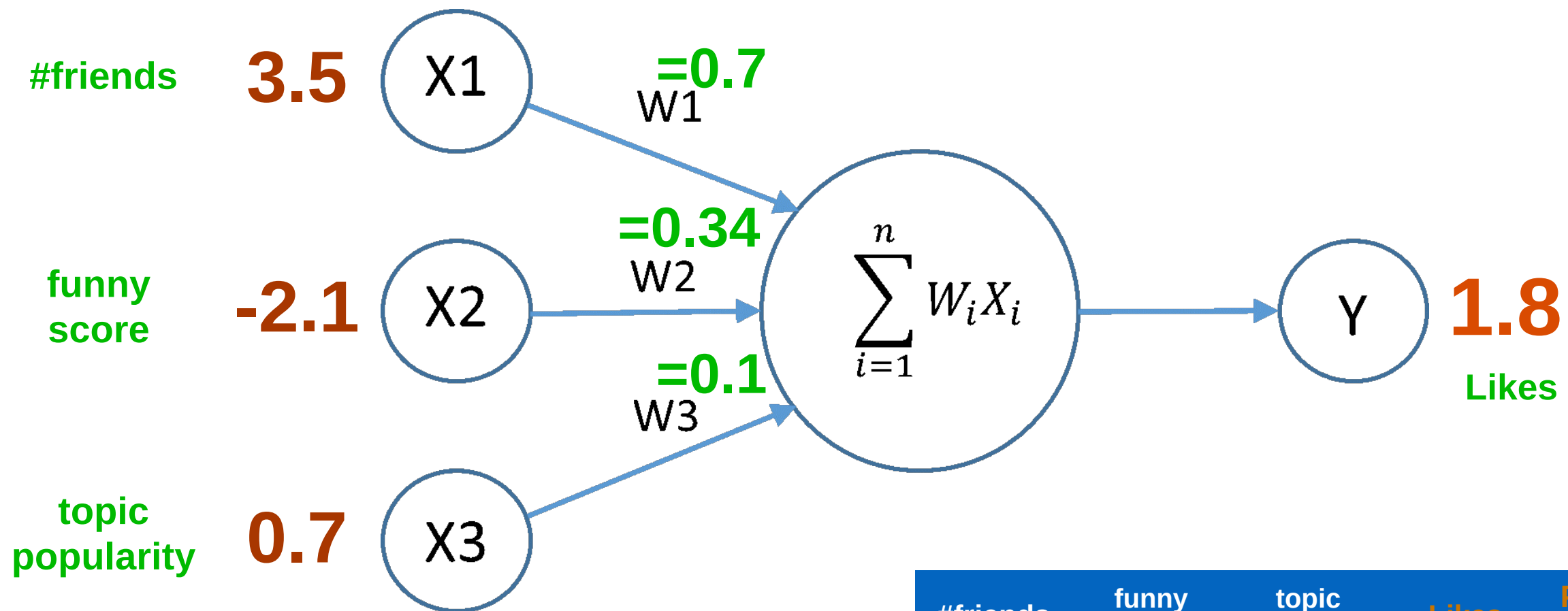


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A Simple Model of a Neuron

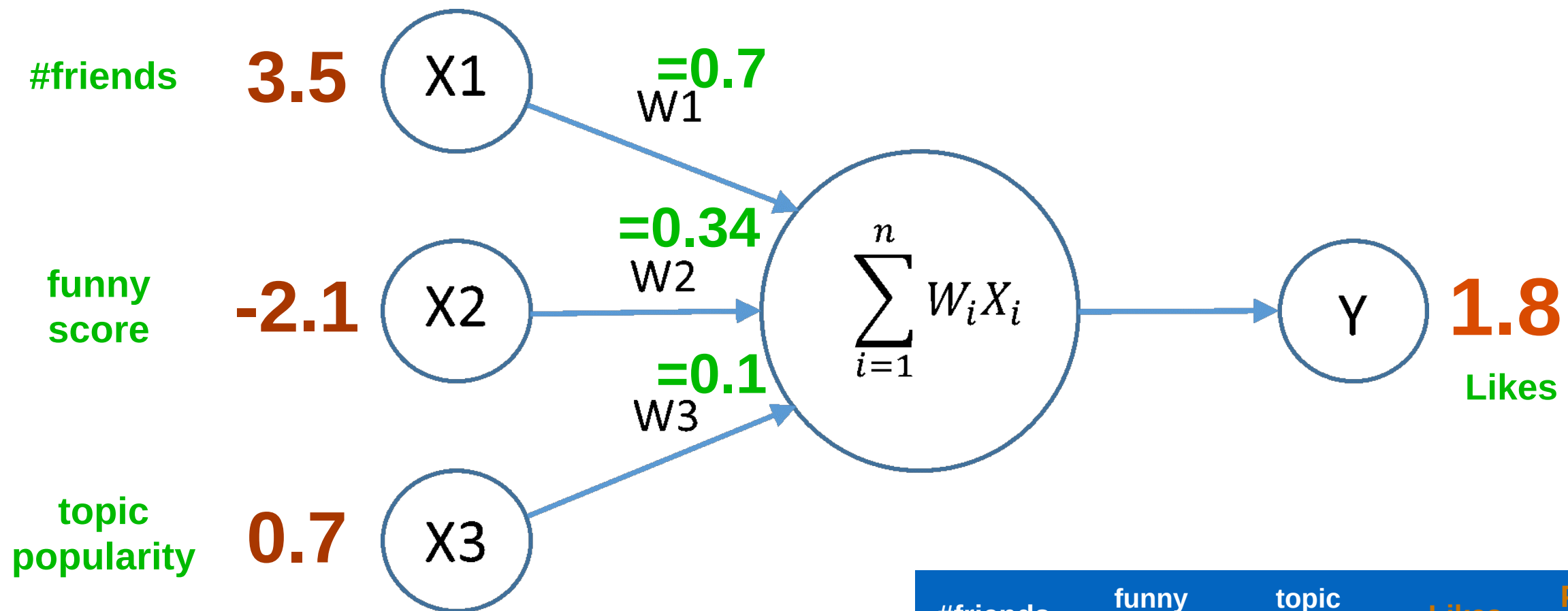


Tests example
#2

Adjusts weights
based on the error
(126-179)

#friends	funny score	topic popularity	Likes	Prediction Likes
1.0	2.4	0.8	25	26.5
45.2	9.1	2.4	126	179
3.5	-2.1	0.7	2.95	
0	113	45	25.6	

A Simple Model of a Neuron

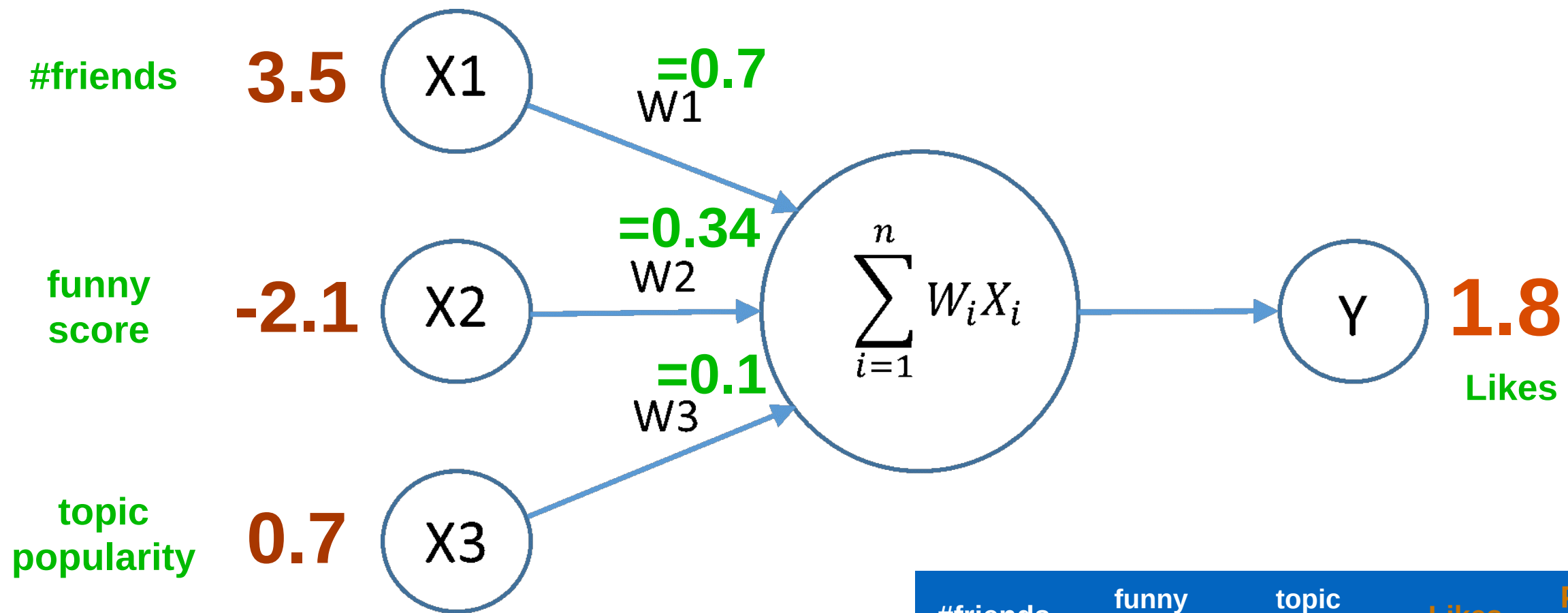


Tests example
#2

Adjusts weights
based on the error
(126-179)

#friends	funny score	topic popularity	Likes	Prediction Likes
1.0	2.4	0.8	25	26.5
45.2	9.1	2.4	126	119
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0	113	45	25.6	

A Simple Model of a Neuron



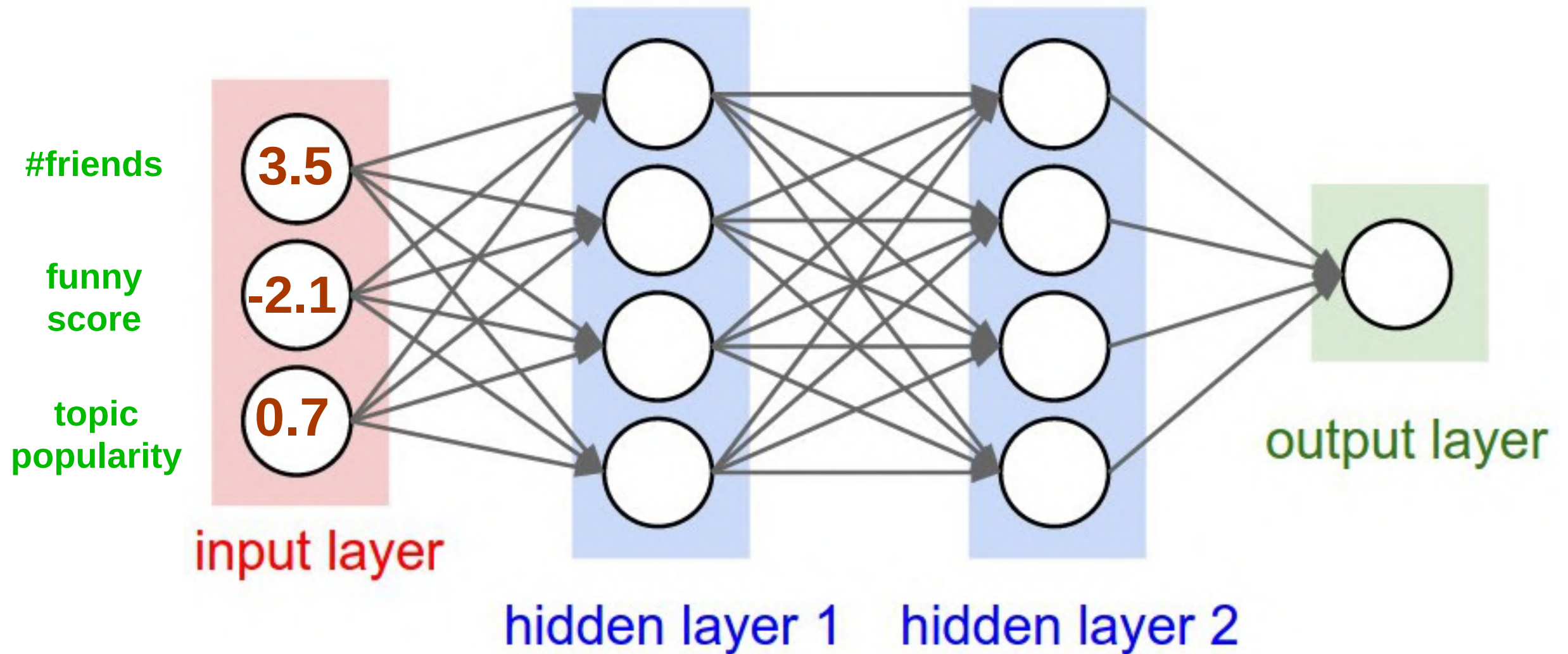
#friends	funny score	topic popularity	Likes	Prediction Likes
1.0	2.4	0.8	25	26.5
45.2	9.1	2.4	126	119
3.5	-2.1	0.7	2.95	1.03
0	113	45	25.6	

Adjusts weights
based on the error
(2.95-1.03)

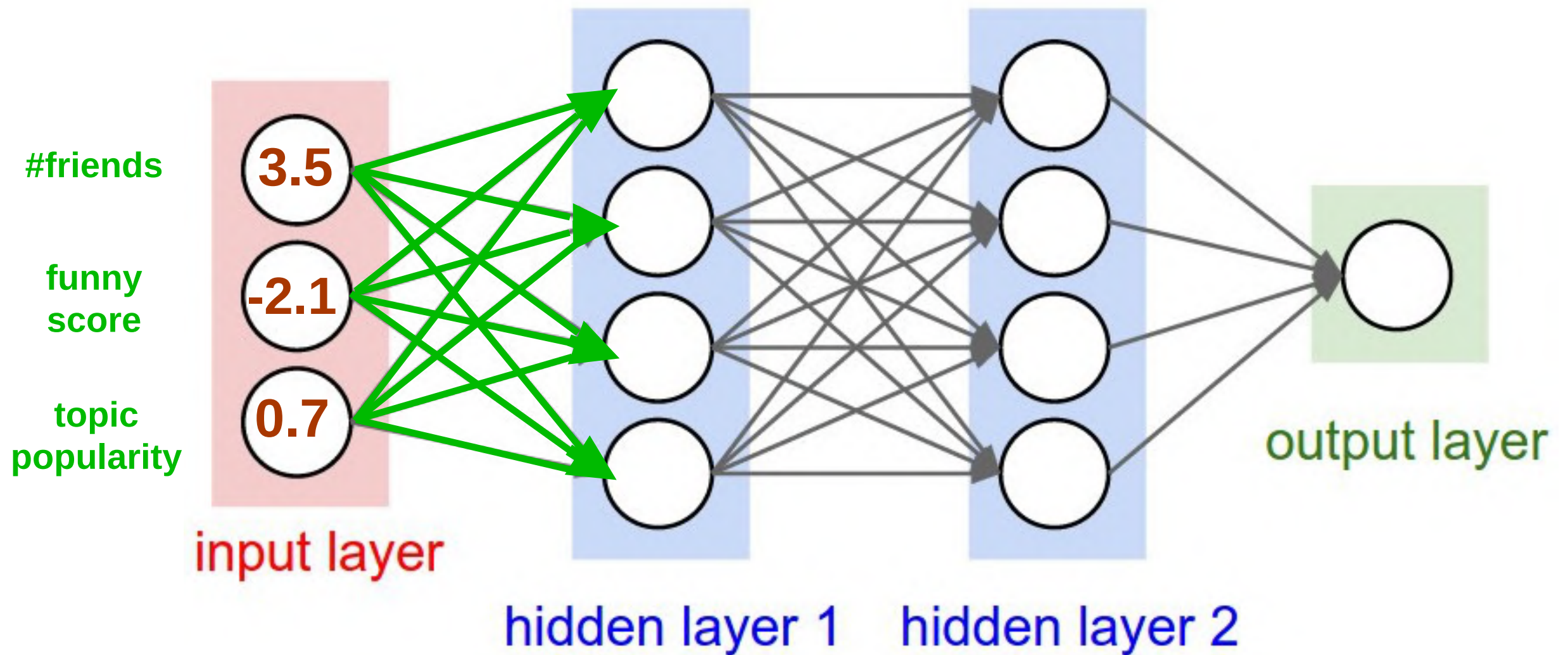
Tests example
#3



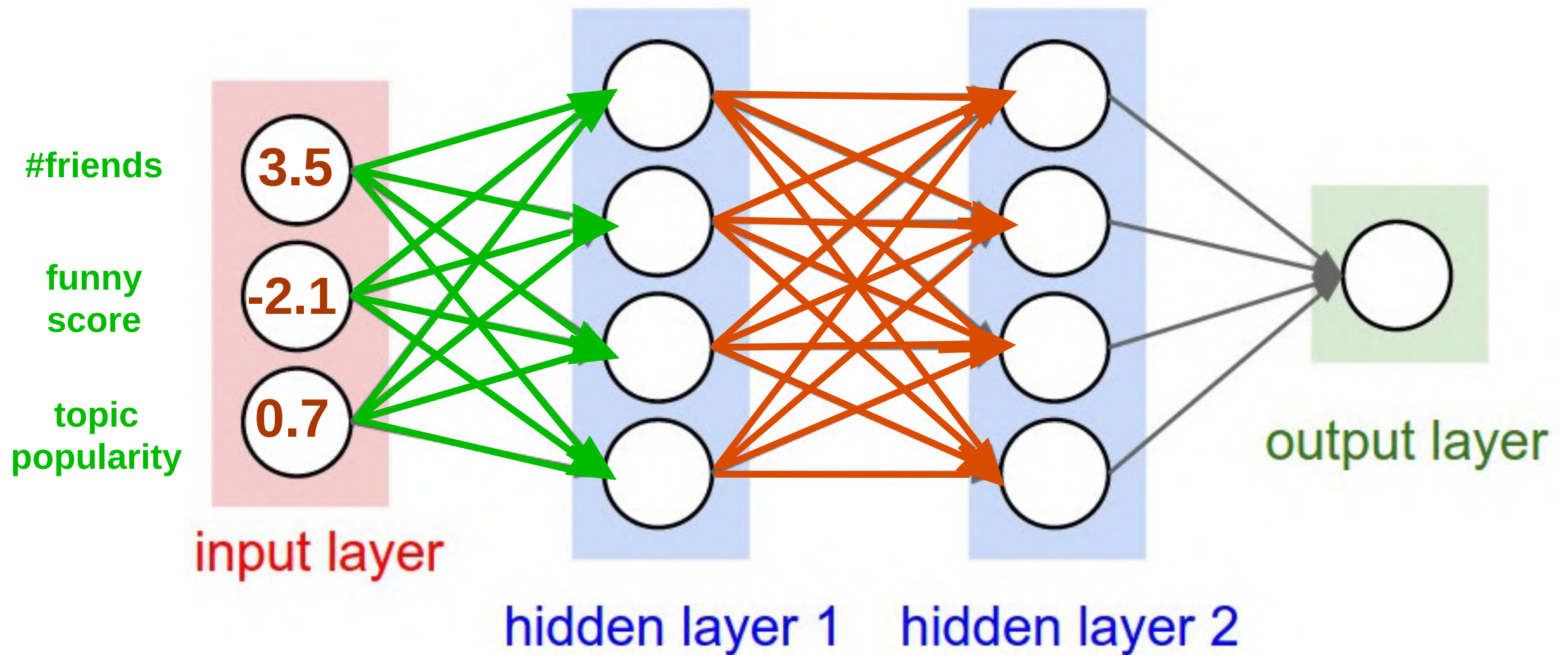
Multi-Layer Neural Networks



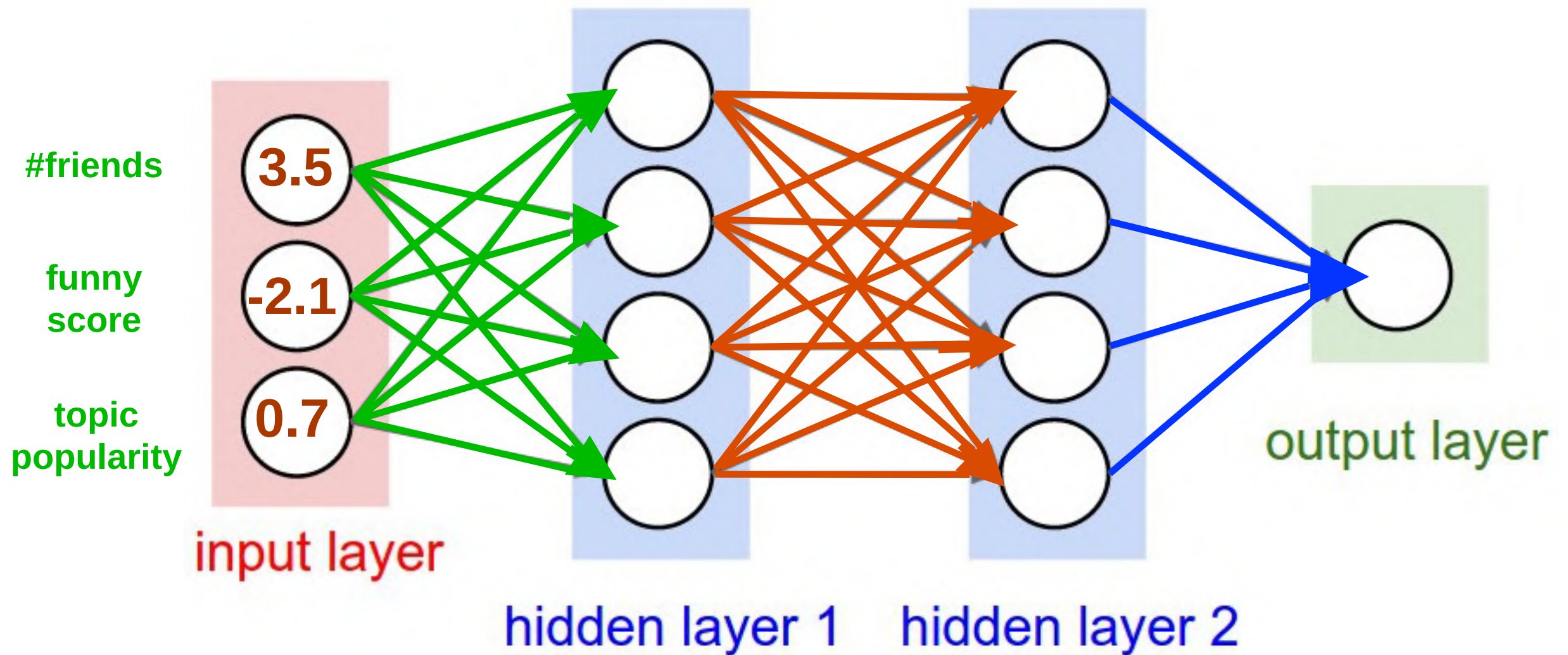
Multi-Layer Neural Networks



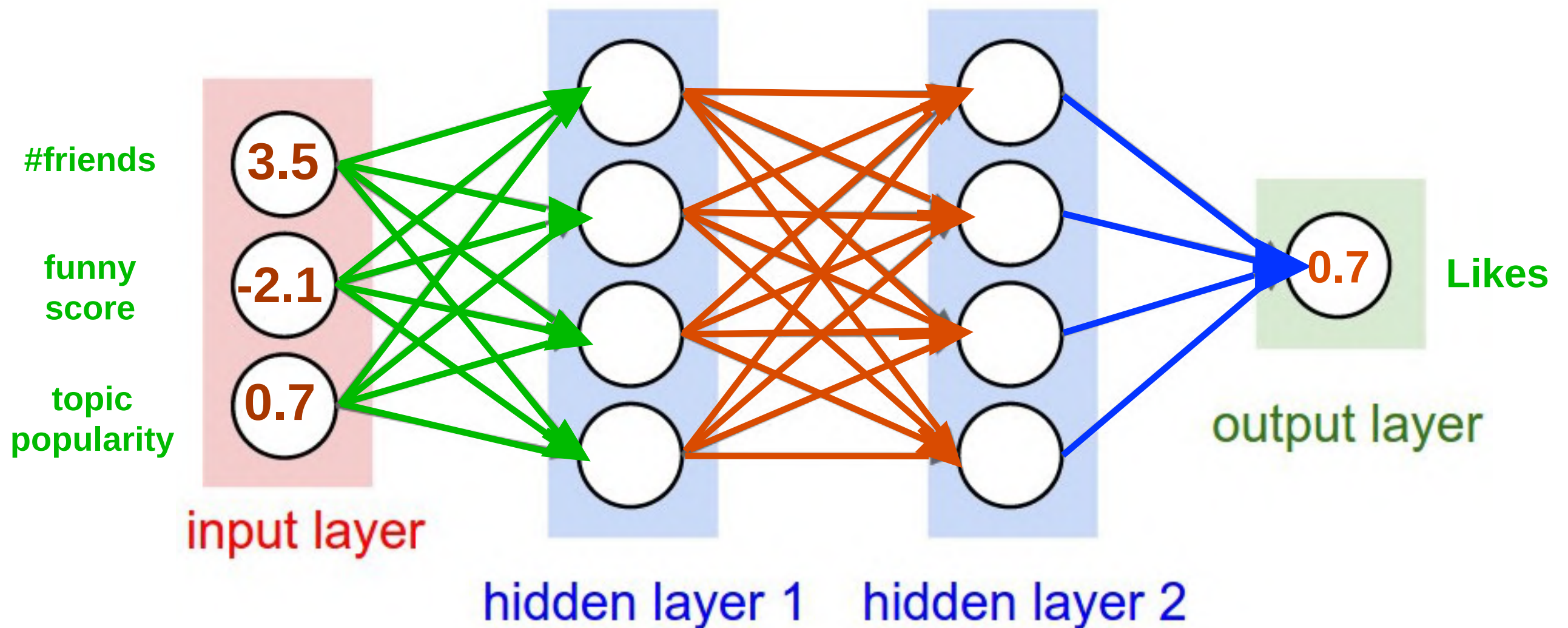
Multi-Layer Neural Networks



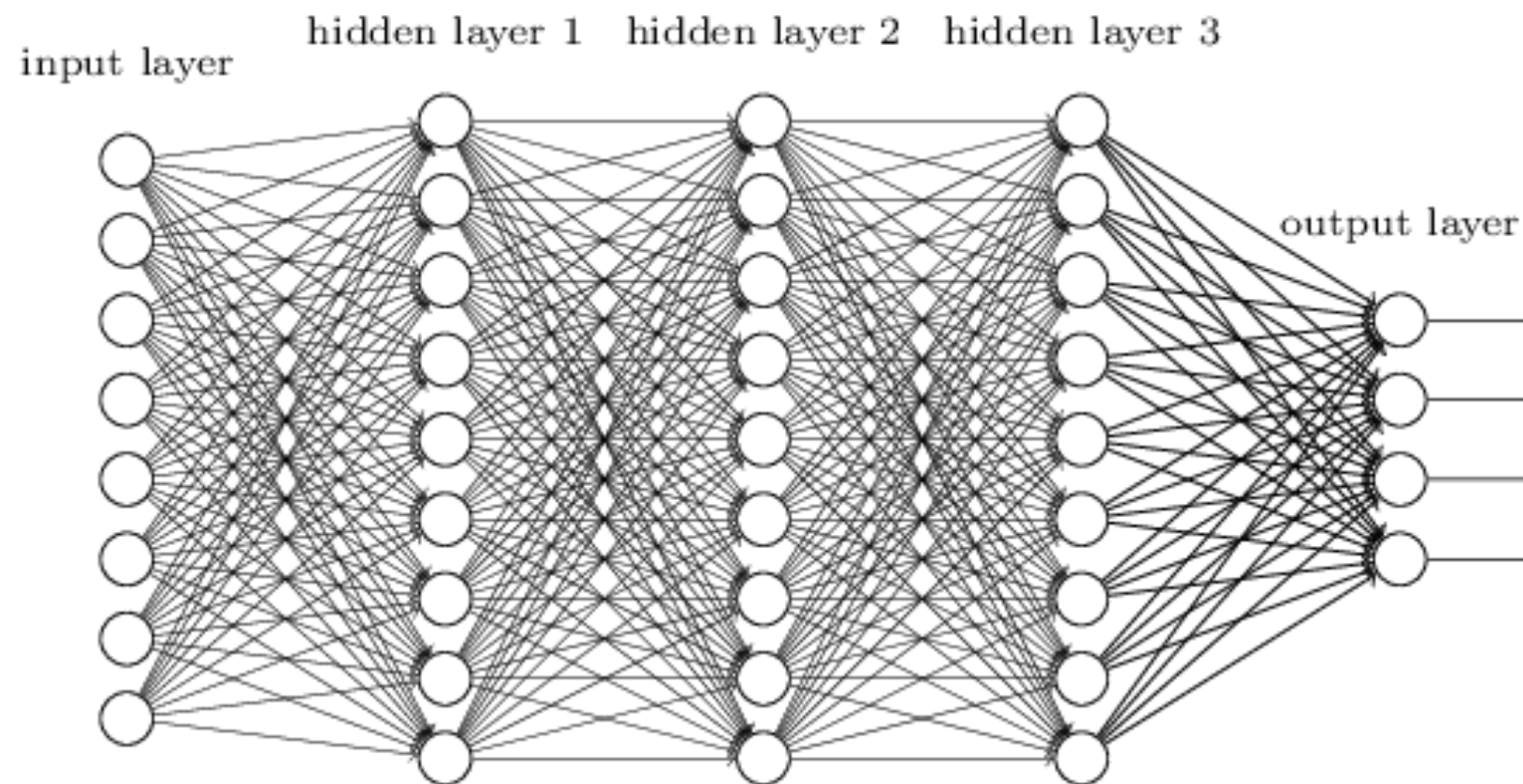
Multi-Layer Neural Networks



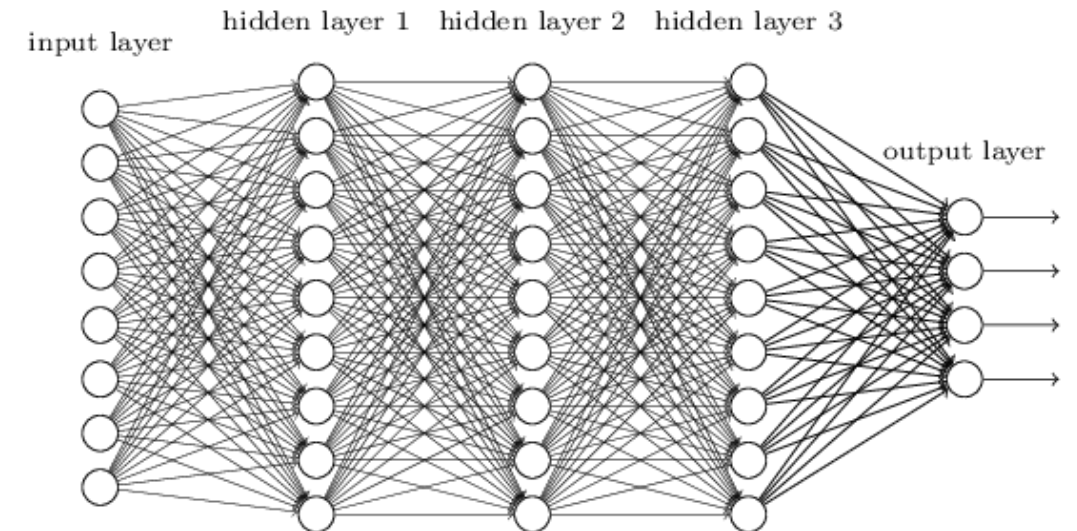
Multi-Layer Neural Networks



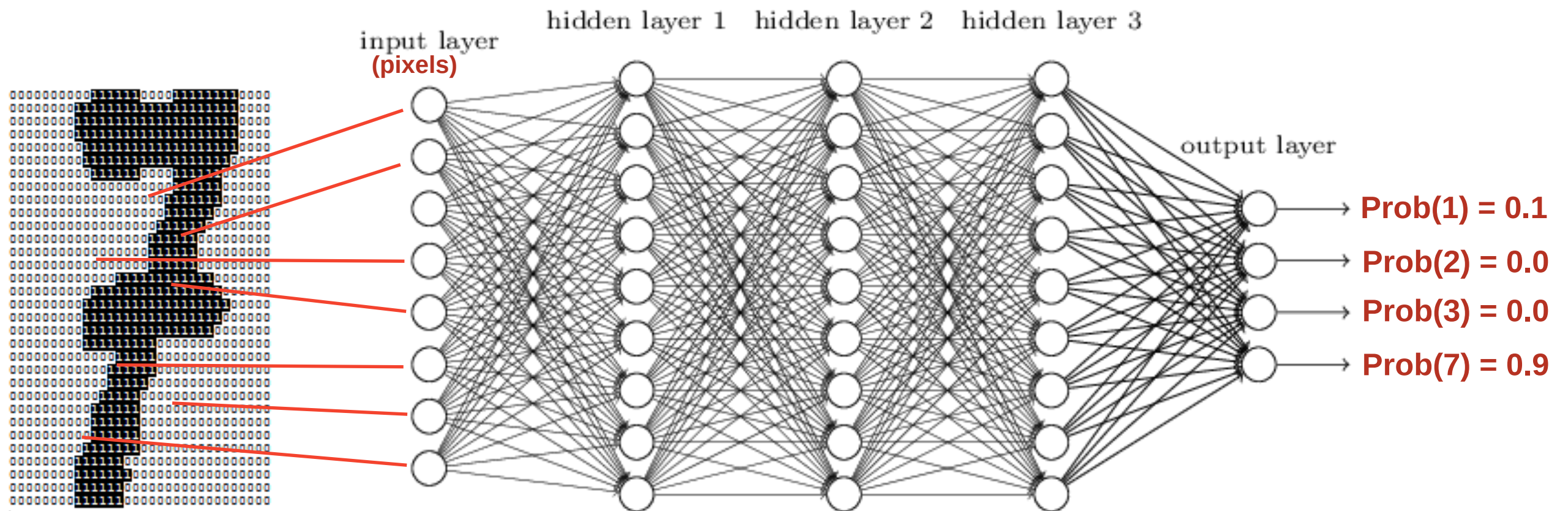
Multi-Layer Neural Networks



Handwriting Recognition



Handwriting Recognition

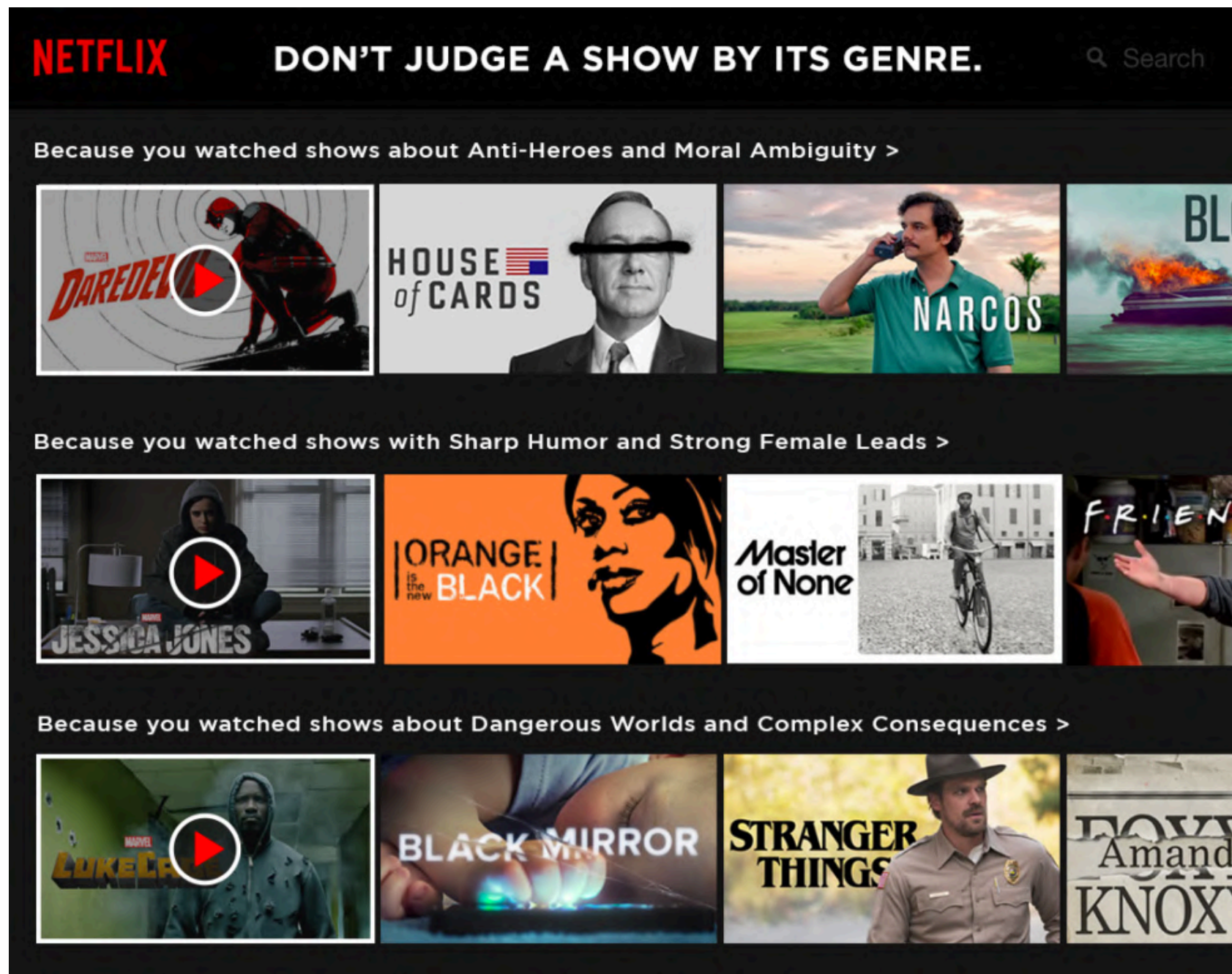


Object Identification in Real-Time



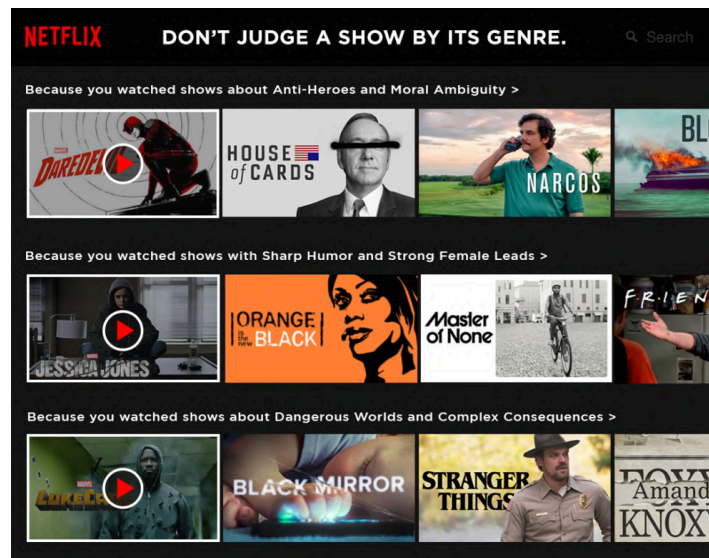
https://www.youtube.com/watch?v=n5uP_LP9SmM

Recommendation Systems

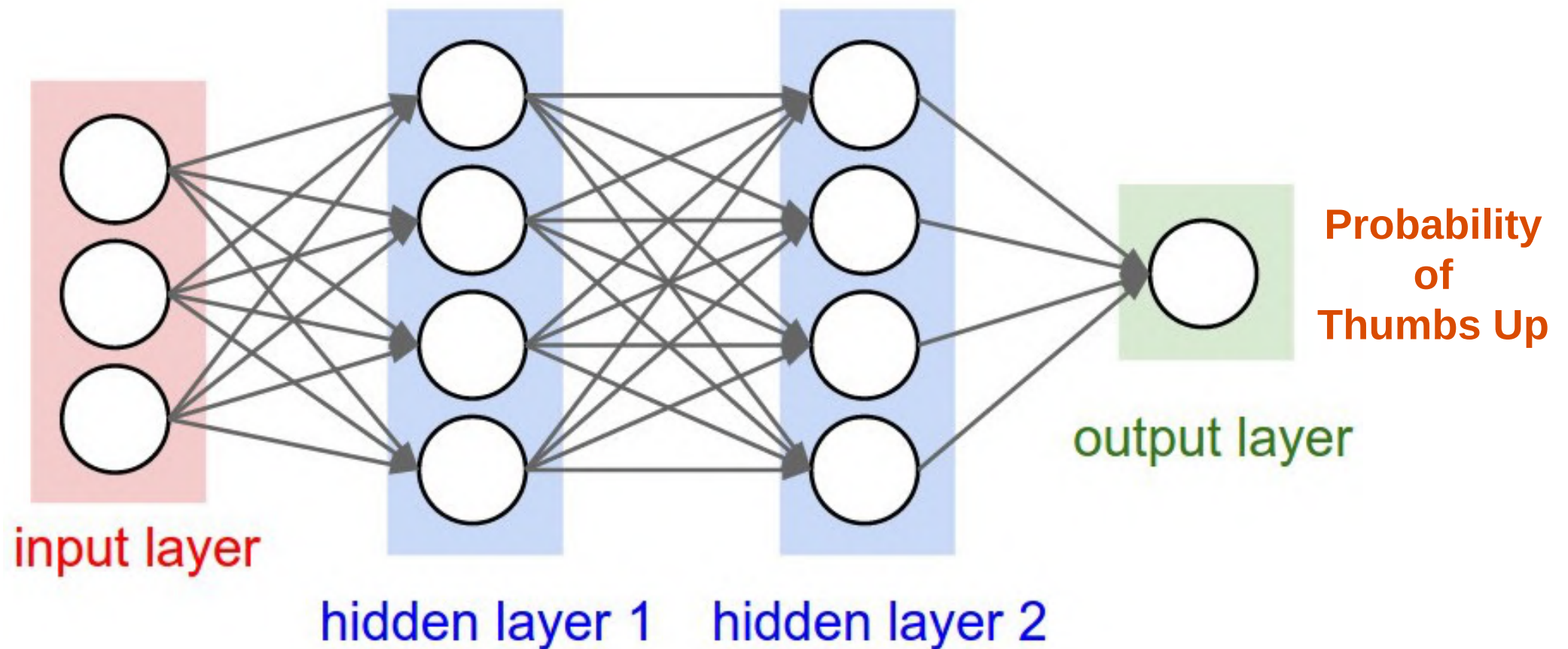


Learning User Preferences & Interests

Recommendation Systems

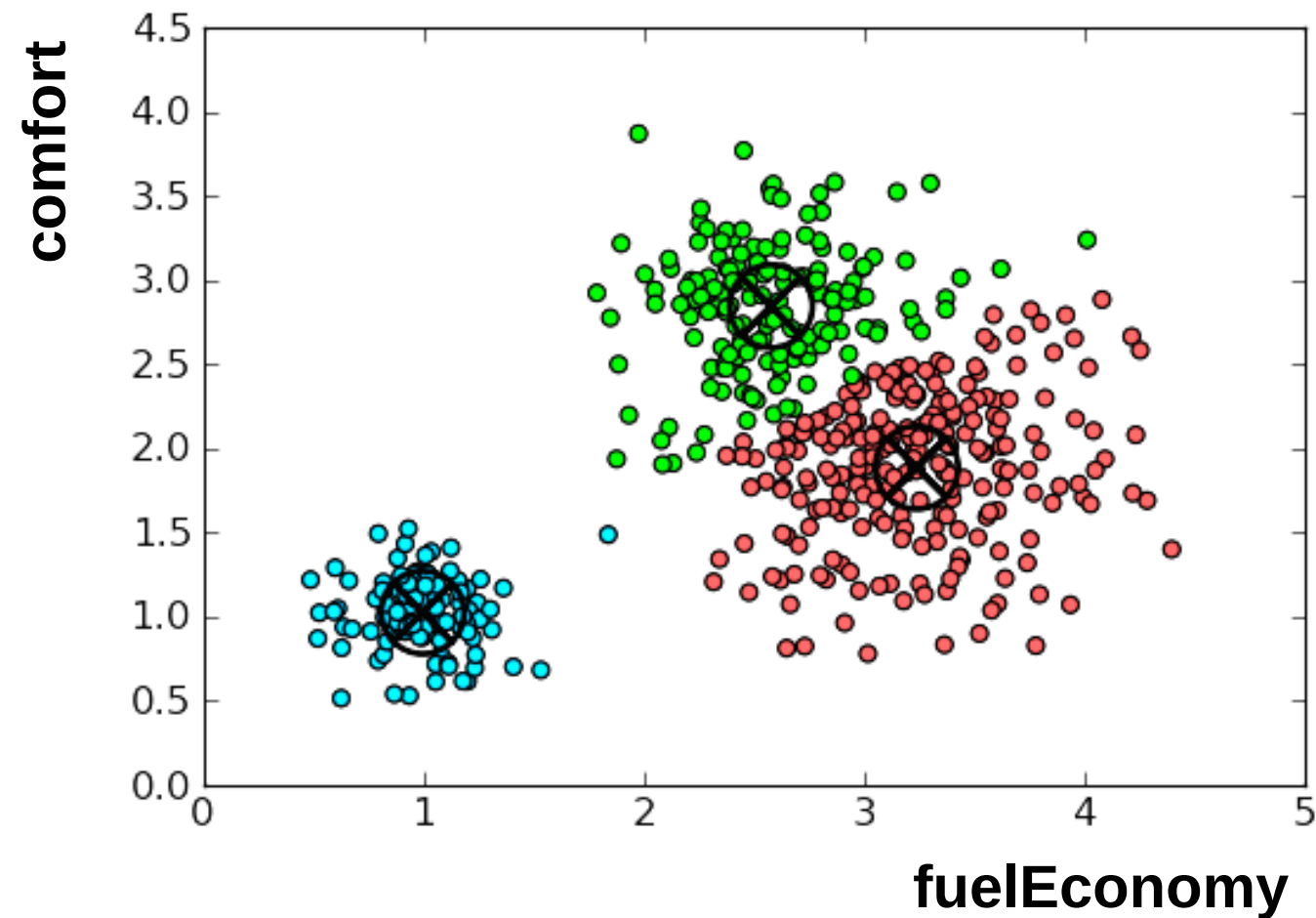


Comedy (0/1)
Strong Female Lead
Year



Unsupervised Learning

Clustering Algorithms



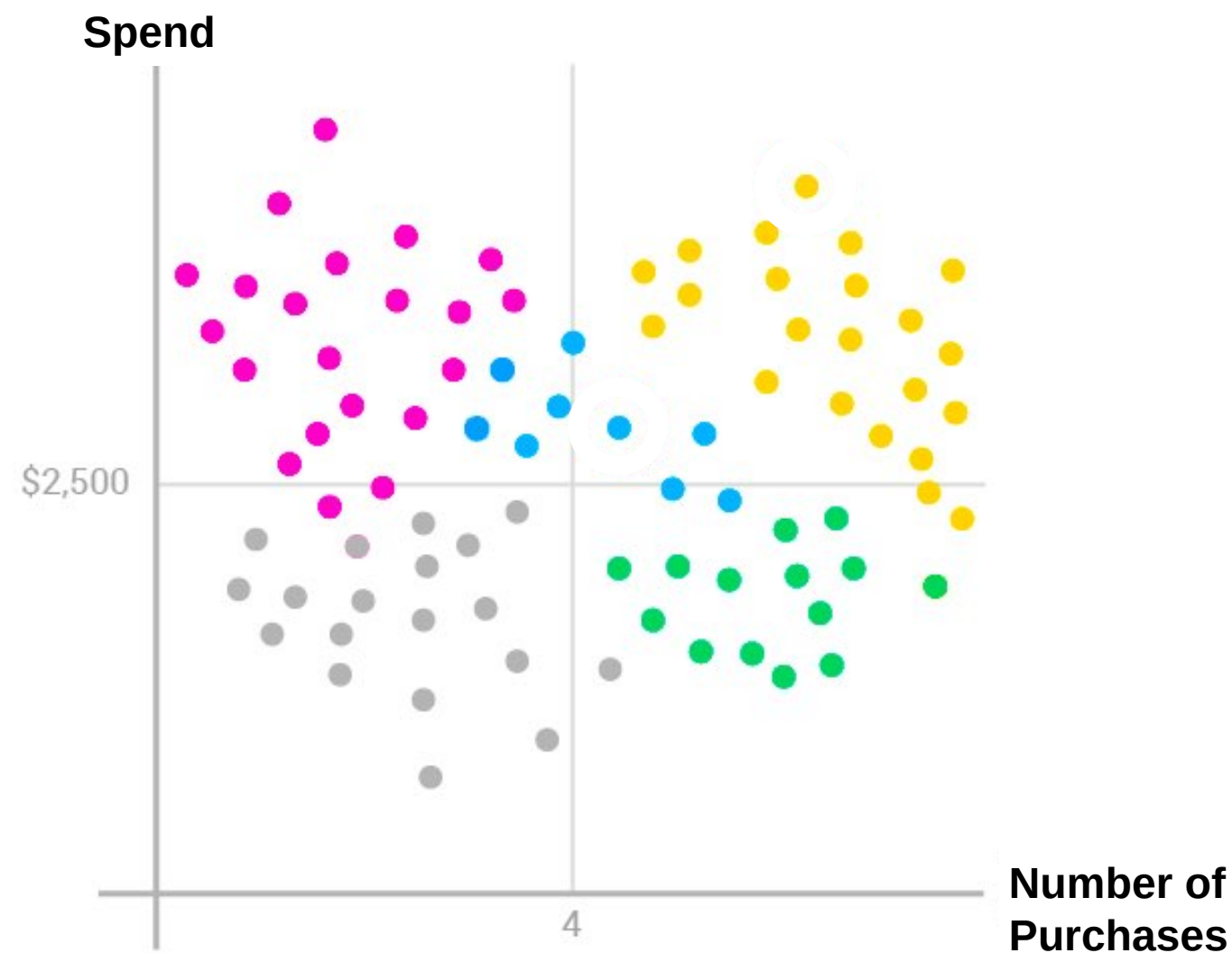
“luxury car buyers”

“car buyers interested in performance/speed”

“car buyers who prioritize economy and comfort”

Clustering Algorithms

- Segmentation of customers based on their behavior
 - directed digital marketing
 - identifies groups (or “types”) of customers based on the way they behave/buy
 - useful to predict which ads might be more relevant to a specific customer



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Customers	Days Since Last Purchase	Number of Purchases (Past 12 Months)	Net Revenue (Past 12 Months)
-----------	--------------------------	--------------------------------------	------------------------------

|— 3 attributes being analyzed —|

Clustering Algorithms

- Segmentation of customers based on their behavior
 - directed digital marketing
 - identifies groups (or “types”) of customers based on the way they behave/buy
 - useful to predict which ads might be more relevant to a specific customer

— identified
4 groups
(or *clusters*)
of customers —

Customers		Days Since Last Purchase	Number of Purchases (Past 12 Months)	Net Revenue (Past 12 Months)
High Spenders	922	9	4	\$154
Mid Spenders	581	54	3	\$121
Risk of Churn	807	192	2	\$70
Low Spenders	1,361	192	2	\$4
	3,671	447	3	\$87

— 3 attributes being analyzed —

Reinforcement Learning (and robotics)



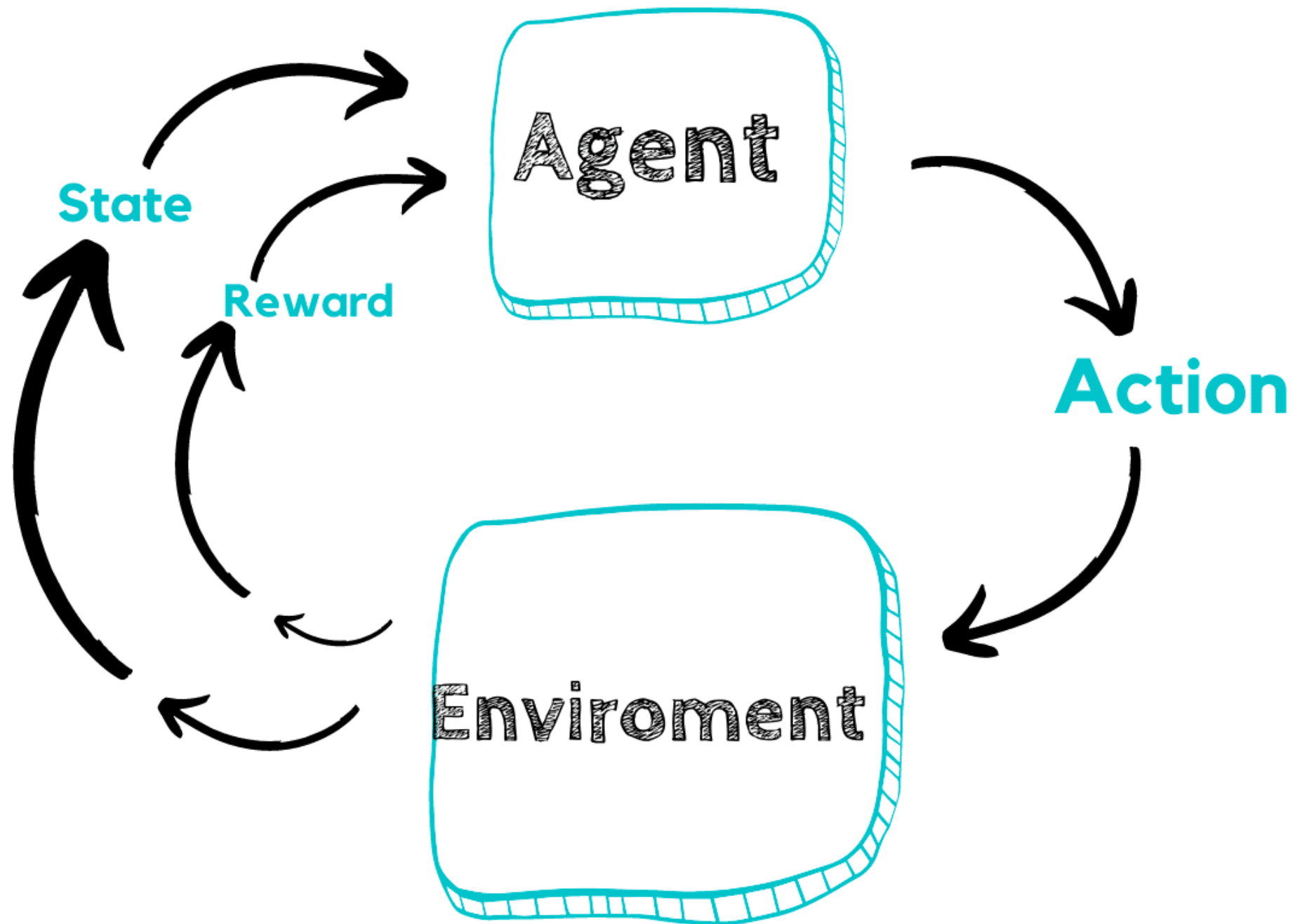
COMPSCI H589!



Reinforcement Learning

- Algorithms to learn how to behave / solve a task / select actions
 - no dataset containing examples of how to behave in different situations
 - no prior knowledge about the environment/system
 - learns only based on rewards (or punishments)
- Positive reinforcement
more likely that behavior will occur again in the future
- Negative reinforcement
less likely that behavior will occur again in the future

Reinforcement Learning



Reinforcement Learning

- Algorithms to learn how to behave / solve a task / select actions
 - no dataset containing examples of how to behave in different situations
 - no prior knowledge about the environment/system
 - learns only based on rewards (or punishments)



States?
Actions?
Rewards?

Reinforcement Learning

Digital Marketing

State: user information

Actions: ad that can be shown

Goal: maximize number of clicks

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AdChoices ▶



Reinforcement Learning

Digital Marketing

State: user information

Actions: ad that can be shown

Goal: maximize number of clicks (reward +1 when user clicks)

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Official Site. Stay Smart. Find great comfort & low prices!
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AdChoices ▶



Reinforcement Learning



<https://www.youtube.com/watch?v=SH3bADiB7uQ>

Reinforcement Learning



Future & Challenges

Impacts of ML on Society



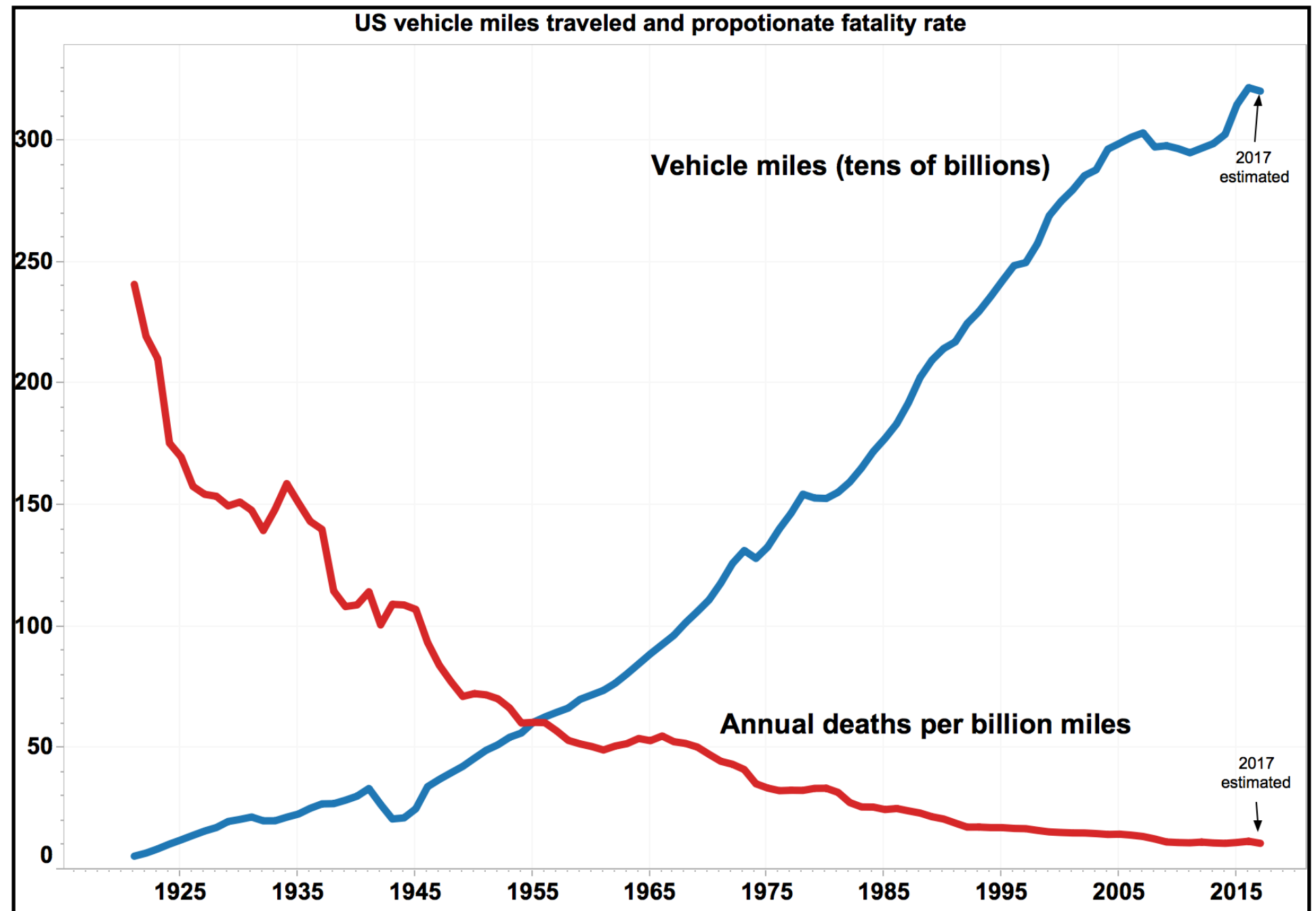
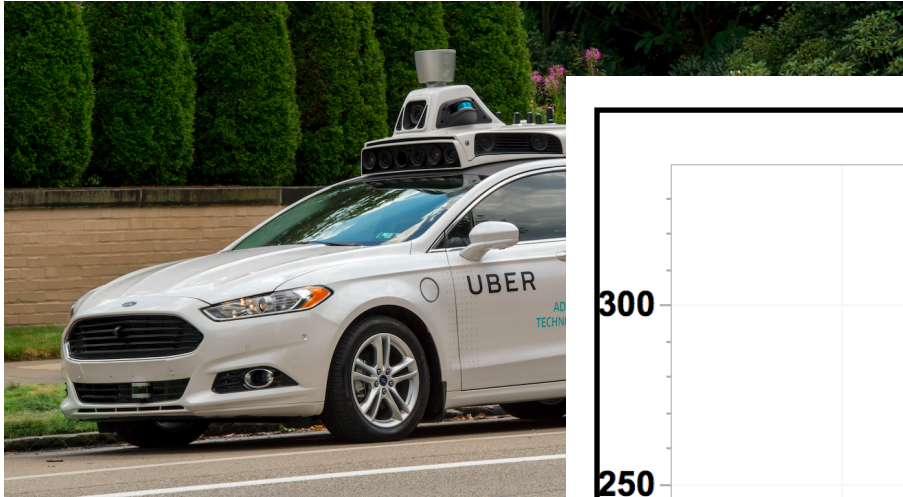
Pros and Cons?

Impacts of ML on Society



“Self-Driving Cars Could Transform Jobs Held by 1 in 9 U.S. Workers”

Impacts of ML on Society



Traffic Fatalities in the United States

Impacts of ML on Society



Good?
Bad?



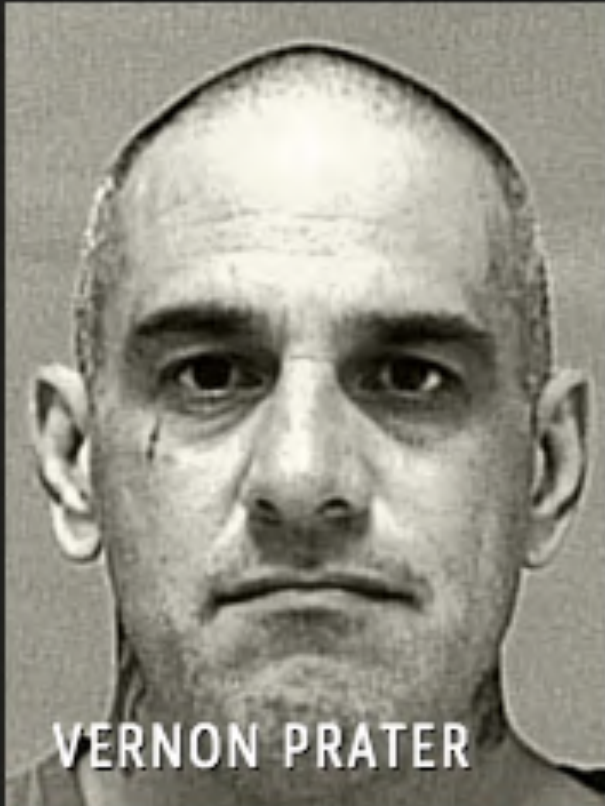
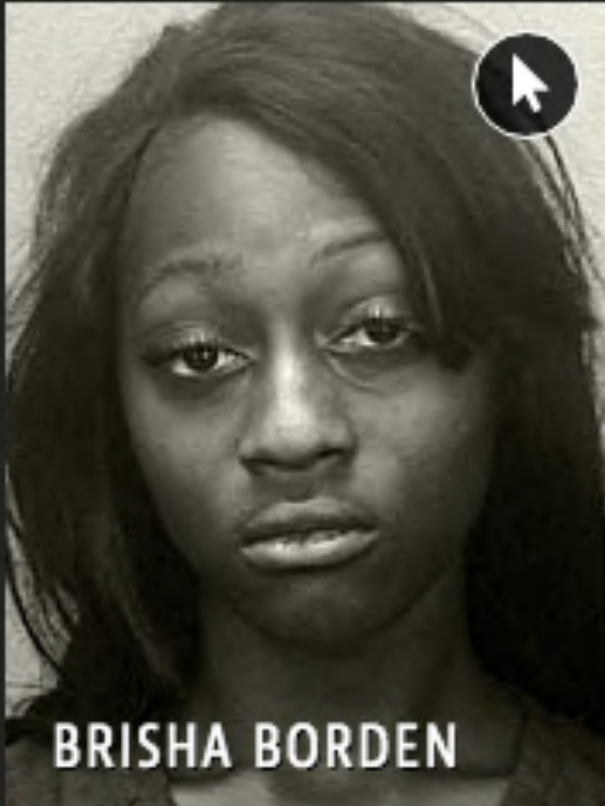
Traffic Fatalities in the United States

Impacts of ML on Society

Crime Prediction



Two Petty Theft Arrests

	
VERNON PRATER	BRISHA BORDEN
LOW RISK 3	HIGH RISK 8

Impacts of ML on Society

Crime Prediction



Two Petty Theft Arrests	
VERNON PRATER	BRISHA BORDEN
Prior Offenses 2 armed robberies, 1 attempted armed robbery	Prior Offenses 4 juvenile misdemeanors
Subsequent Offenses 1 grand theft	Subsequent Offenses None
LOW RISK 3	HIGH RISK 8

“black defendants (...) were twice as likely to be misclassified as [future criminals]”

Impacts of ML on Society

Crime Prediction



Two Petty Theft Arrests	
VERNON PRATER	BRISHA BORDEN
Prior Offenses 2 armed robberies, 1 attempted armed robbery	Prior Offenses 4 juvenile misdemeanors
Subsequent Offenses 1 grand theft	Subsequent Offenses None
LOW RISK 3	HIGH RISK 8

Machine learning algorithms can (and do!) discriminate

Impacts of ML on Society



Automation in the workplace

Still a long way to go...





Machine Learning

CMPSCI 589

Bruno C. da Silva

bsilva@cs.umass.edu

An Introduction to ML

UMassAmherst
College of Information
& Computer Sciences



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